Supplementary Information

How does complex knowledge promote regional economic growth within the "Belt and Road"?

SIN-SOM (SERGIO) TSIONG†

School of International and Public Affairs, Shanghai Jiao Tong University, China.

SHUAIHANG LI

College of Business, Shanghai University of Finance and Economics, Shanghai, China.

MINGQIAN ZHANG

School of Economics and Finance, Shanghai International Studies University, Shanghai, China.

†First and corresponding author: Sin-Som (Sergio) Tsiong. E-mail: s.sergio.zhang@gmail.com; ORCID: 0000-0003-3271-3796; Address: No. 1954, Huashan Rd., Xuhui District, Shanghai, China.

Second author: Shuaihang Li. E-mail: lsh1177@126.com; Address: No.100 Wudong Rd., Yangpu District, Shanghai, China.

Third author: Mingqian Zhang. E-mail:E-mail: zhangmq@shisu.edu.cn; Address: No. 550, Dalian Rd. (W), Hongkou District, Shanghai, China.

Contents

List of Tables

List of Figures

Appendix A Knowledge Complexity Measurement

A.1 Bipartite Network Framework

In research on international trade, the bipartite network can map the trade information of different countries on different products, doing well in clarifying the underlying relationships among countries or among products, which cannot be realized by traditional complex networks (Hidalgo and Hausmann, 2009; Sciarra et al., 2020; Tacchella et al., 2013). We can construct a country-product bipartite network (the Adjacent Matrix corresponding is M_{ct}) by mapping information of patent outputting of different countries on different technology domains, based on defining the two different types of vertexes: countries and products, and edges between two different types of vertexes.

Within the framework of the bipartite network, issues on complexity can be divided into two dimensions, namely, discussing the complexity of two objects corresponding to the two types of vertexes. For the framework constructed in this article, issues of knowledge complexity can be described by determining two properties — X_c and Y_t, which represent the complexity properties of counties and products, respectively. The property X_c can be expressed as a function composed of property Y_1,Y_2,...,Y_t and the Adjacency Matrix M_{ct} of the bipartite network; at the same time, property Y_t can be also expressed as a function composed of property X_1,X_2,...,X_c and M_{ct}. And then, studying economic problems is just equal to solving the coupled equations below:

$$\begin{cases} X_c = f(Y_1, Y_2, \ldots, Y_t, M_{ct}) \\ Y_t = g(X_1, X_2, \ldots, X_c, M_{ct}) \end{cases} \tag{S.1}$$

The two functions in Eq.(S.1) allows one to recast the determination of X_c and Y_t as the solutions of an eigen-problem of a suitable transformation matrix $\boldsymbol{W}$.

$$\begin{cases} X_c = \frac{1}{\sqrt{\lambda}} \sum_t W_{ct} Y_t \\ Y_t = \frac{1}{\sqrt{\lambda}} \sum_c W_{ct} X_c \end{cases} \Leftrightarrow \begin{cases} X_c = \frac{1}{\lambda} \sum_t \sum_{c'} W_{ct} W_{c't} X_{c'} = \frac{1}{\lambda} \sum_{c'} N_{cc'} X_{c'} \\ Y_t = \frac{1}{\lambda} \sum_c \sum_{t'} W_{ct} W_{ct'} Y_{t'} = \frac{1}{\lambda} \sum_{t'} G_{tt'} Y_{t'} \end{cases} \tag{S.2}$$

Where, W_{ct} can be calculated from M_{ct} by some kind of algorithm; λ is eigenvalue. $N_{cc'} = (\boldsymbol{W}\boldsymbol{W}^T)_{cc'}$ and $G_{tt'} = (\boldsymbol{W}^T\boldsymbol{W})_{tt'}$ are symmetrical square matrixes; thereinto, $\boldsymbol{N}$ can be explained as the proximity matrix of the country, describing the similarity of the structure of countries' participating in technology domains; $\boldsymbol{G}$ can be correspondingly explained as the proximity matrix of the technology, describing the similarity of the structure of technology domains' being participated by countries.

A.2 The Fitness and Complexity algorithm

The coupled equations Eq.(S.2) are systemically studied by Hidalgo and Hausmann (2009), they proposed the MR in calculation the transformation matrix W. But, as mentioned above, the flaw of information omission is essentially rooted in the MR, which means that the MR may conduct incorrect evaluations on the diversification of countries and ubiquity of products (Hausmann et al., 2014; Morrison et al., 2017); what's more, the non-convergency in the iterations may also have impacts on calculation results significantly (Pugliese et al., 2016). That implies that, while measuring the complexity based on the MR within the bipartite network, the complexity property, of one type of vertex, may not be sensitive enough to changes on the same property of the other type, and the MR cannot reflect fine structure of the pattern on interaction between two types of vertexes.

Actually, there should be a difference when discussing the complexity of these two types of vertexes, which can be reflected in the difference in their own formulas. While referring to a certain country, the wider the technology domains it participates in, the higher the knowledge complexity it will be; in addition, when the centrality or importance of a technology the country participating in, and more deeply the country participating in such technology, the higher the knowledge complexity it will be. As for a certain technology, the situation is different. When the participating countries of the technology are more important, the knowledge complexity of the technology is higher; but, if it is more widely participated by countries, namely the number of participating countries is larger, the knowledge complexity should be lowered.

Tacchella et al. (2013) proposed the Fitness and Complexity (FC) algorithm, which expresses the interrelationships between two types of vertexes as non-linear functions. This algorithm can not only acquire almost all information calculated from the MR (Mariani et al., 2015; Sciarra et al., 2020), but also avoid the shortcomings of the MR mentioned above (Kemp-Benedict, 2014; Morrison et al., 2017). Therefore, this article will adopt the FC algorithm, and recast the mathematical expression of the complexity issues from Eq.(S.2), by determining two complexities: F_c — fitness for countries, and Q_t — complexity for technologies:

$$\begin{cases} \widetilde{F_c}^{(0)} = \sum_t M_{ct} {Q_t}^{(n-1)}, & \forall c \\ \widetilde{Q_t}^{(0)} = \frac{1}{\sum_c M_{ct} \frac{1}{{F_c}^{(n-1)}}}, & \forall t \end{cases} \Leftrightarrow \begin{cases} {F_c}^{(n)} = \frac{\widetilde{F_c}^{(n)}}{\sum_c \widetilde{F_c}^{(n)}/C} \\ {Q_t}^{(n)} = \frac{\widetilde{Q_t}^{(n)}}{\sum_t \widetilde{Q_t}^{(n)}/T} \end{cases} \tag{S.3}$$

and the initial conditions are: $\begin{cases} \widetilde{F_c}^{(0)} = 1, & \forall c \\ \widetilde{Q_t}^{(0)} = 1, & \forall t \end{cases}$.

In Eq.(S.3), the intermediations of fitness and complexity, $\widetilde{F_c}^{(n)}$ and $\widetilde{Q_t}^{(n)}$, are all standardized in each step, which make $\widetilde{F_c}^{(n)}$ and $\widetilde{Q_t}^{(n)}$ tend to be converged. Therefore, the equations in iterative form can be expressed as non-iterative form, by introducing rescaling factors $c_F = {}^{C}/_{\sum_t Q_t s_t}$ and $c_Q = {}^{\sum_t Q_t s_t}/_{T}$:

$$\begin{cases} F_c = c_F \sum_t M_{ct} Q_t \\ Q_t = c_Q \frac{1}{\sum_c M_{ct} \frac{1}{F_c}} \end{cases} \tag{S.4}$$

where s_t is the weight of technology vertex, namely $s_t = \sum_c M_{ct}$. In Eq.(S.4), Q_t is expressed as a non-linear function $f_{non-linear}(F_1, F_2, \ldots, F_c, M_{ct})$. Based on the Taylor Expansion (Sciarra et al., 2020), Eq.(S.4) can be approximatively expressed in a linear form:

$$\begin{cases} F_c \backsimeq c_F \sum_t M_{ct} Q_t \\ Q_t \backsimeq \frac{c_Q}{(s_t')^2} \sum_c \frac{M_{ct} F_c}{{s_c}^2} \end{cases} \tag{S.5}$$

where s_c is the weight of country vertex, namely $s_c = \sum_t M_{ct}$ and $s_t' = \sum_c {}^{M_{ct}}/_{s_c}$.

Eq.(S.5) is a linear form of Eq.(S.1) based on the FC algorithm, let $\begin{cases} X_c = {}^{F_c}/_{s_c} \\ Y_t = {}^{Q_t}/_{s_{t'}} \end{cases}$, and

$\boldsymbol{N}/\boldsymbol{G}$ can be expressed as:

$$\begin{cases} N_{cc'} = \frac{\sum_{c'} M_{ct} M_{c't}}{(s'_t)^2 s_{c'} s_c} \\ G_{tt'} = \frac{\sum_{t'} M_{ct} M_{ct'}}{(s_t)^2 s'_{t'} s'_t} \end{cases} \tag{S.6}$$

A.3 The matrix-estimation exercise

While conducting research on international trade within the bipartite network, the FC algorithm, improved by the MR, provides a feasible solution for the measurement of centrality, which has been used in studies on complexity gradually. Notwithstanding that, it is inevitable for either the MR or the FC algorithm to generate ad hoc assumptions for their purpose of calculating the Eigenvector Centrality to some extent, which will make the results lack effectiveness Sciarra et al. (2018).

Thus, based on combining the FC algorithm and the matrix-estimation exercise, this article constructs indicators for measuring the knowledge complexity of countries and technologies by solving the eigen-problem described by Eq.(2), which are GPYC (Generalized comPlexitY of Country) and GPYT (Generalized comPlexitY of Technology) respectively.

Consider adopting the methods of matrix-estimation exercise. In terms of an adjacency matrix $\boldsymbol{A}$, the matrix estimator $\hat{A}_{ij}$ will depend on the centrality of vertex i and j, namely:

$$\hat{A}_{ij} = f(x_i, x_j) \tag{S.7}$$

The same as OLS, $\hat{A}_{ij}$ can be estimated by minimizing its Squared Error (SE). Actually, SE be partitioned into SE_0 and SE_k; thereinto SE_0 is independent of x_k, and SE_k depending on x_k:

$$SE = SE_0 + SE_k = \sum_i \sum_j \left(A_{ij} - \hat{A}_{ij}\right)^2 = \sum_i \sum_j \left(A_{ij} - f(x_i, x_j)\right)^2 \tag{S.8}$$

Because of SE_0 being independent of x_k, minimizing SE just equal to minimizing SE_k, which means that the derivate of SE_k should be equal to 0, namely:

$$\begin{aligned} \frac{\partial SE}{\partial x_k} &= \frac{\partial SE_k}{\partial x_k} \\ &= 4\sum_{i\neq k} \left(A_{ik} - f\left(x_i, x_k\right)\right) \frac{\partial f\left(x_i, x_k\right)}{\partial x_k} + 2\left(A_{kk} - f\left(x_k, x_k\right)\right) \frac{\partial f\left(x_k, x_k\right)}{\partial x_k} = 0 \end{aligned} \tag{S.9}$$

And based on the concept of unique contribution of commonality analysis, the unique contribution of the vertex k can be expressed as:

$$UC_k = R_N^2 - R_{N\backslash k}^2 = \frac{SE_{N\backslash k} - SE_N}{\sum_i \sum_j A_{ij}(1 - \frac{\sum_i \sum_j A_{ij}}{N^2})} \tag{S.10}$$

where R_N^2 denotes the goodness of the estimation $\hat{A}_{ij}$ by considering all centrality of vertexes, and $R_{N\backslash k}^2$ denotes that by excluding vertex k. Since $\sum_i \sum_j A_{ij}(1 - \frac{\sum_i \sum_j A_{ij}}{N^2})$ whether excluding vertex k or not, UC_k just depends on $\Delta SE = SE_{N\backslash k} - SE_N$:

$$\begin{aligned} \Delta SE = & 2\sum_{i\neq k} (f(x_i, 0) - f(x_i, x_k))(f(x_i, 0) + f(x_i, x_k) - 2A_{ik}) \\ & + (f(0,0) - f(x_k, x_k))(f(0,0) + f(x_k, x_k) - 2A_{kk}) \end{aligned} \tag{S.11}$$

Since coupling Equations of Eq.(S.2) refers to eigen problem, given multi-component eigenvector centrality, estimation $\hat{A}_{ij}$ can be re-written as:

$$\hat{A}_{ij}(s) = f(x_i, x_j) = \sum_{t=1}^{s} \gamma_t x_{i,t} x_{j,t} = \gamma_1 x_{i,1} x_{j,1} + \gamma_2 x_{i,2} x_{j,2} + \ldots + \gamma_s x_{i,s} x_{j,s} \tag{S.12}$$

Then, ΔSE can be expressed as:

$$\Delta SE(s) = -2 \sum_{t=1}^{s} \gamma_t^2 x_{k,t}^2 \sum_i x_{i,t}^2 + 4 \sum_{t=1}^{s} \gamma_t x_{k,t} \sum_i A_{ik} x_{i,t} + \left(\sum_{t=1}^{s} \gamma_t x_{k,t}^2 \right)^2 \tag{S.13}$$

The unique contribution UC_k of vertex k can be expressed as:

$$UC_k(s) = 2 \sum_{t=1}^{s} \gamma_t^2 x_{k,t}^2 + \left(\sum_{t=1}^{s} \gamma_t x_{k,t}^2 \right)^2 \tag{S.14}$$

Based on the idea of unique contribution, and employ 2 eigenvectors as Sciarra et al. (2018) did, namely set $s = 2$, we construct indicators for measuring the knowledge complexity of countries and technologies, which are GPYC (Generalized comPlexitY of Country) and GPYT (Generalized comPlexitY of Technology) respectively:

$$\begin{cases} GPYC_c = \left(\sum_{i=1}^{2} \lambda_i^N \left(v_{c,i}^N \right)^2 \right)^2 + 2 \sum_{i=1}^{2} \left(\lambda_i^N \right)^2 \left(v_{c,i}^N \right)^2 \\ GPYT_t = \left(\sum_{i=1}^{2} \lambda_i^G \left(v_{t,i}^G \right)^2 \right)^2 + 2 \sum_{i=1}^{2} \left(\lambda_i^G \right)^2 \left(v_{t,i}^G \right)^2 \end{cases} \tag{S.15}$$

Thereinto, λ_1^N/λ_2^N and λ_1^G/λ_2^G are the largest 2 eigenvalues of the proximity matrixes $\boldsymbol{N}$ and $\boldsymbol{G}$, $v_{c,1}^N/v_{c,2}^N$ and $v_{t,1}^G/v_{t,2}^G$ are the eigenvectors corresponding to λ_1^N/λ_2^N and λ_1^G/λ_2^G respectively. Worthing mentioning, $\boldsymbol{N}$ and $\boldsymbol{G}$ here are the modified matrix, by setting all diagonal elements to an arbitrary constant value (such as 0), namely $\begin{cases} N_{cc'} = \frac{\sum_{c'} M_{ct} M_{c't}}{(s_t')^2 s_{c'} s_c}, & \forall c' \neq c \\ N_{cc'} = 0, & c' = c \end{cases}$ and $\begin{cases} G_{tt'} = \frac{\sum_{t'} M_{ct} M_{ct'}}{(s_t)^2 s_{t'}' s_t'}, & \forall t' \neq t \\ G_{tt'} = 0, & t' = t \end{cases}$, and redundant self-proximity can be deleted, making the modified $\boldsymbol{N}$ and $\boldsymbol{G}$ just reflect the inter-proximity information of vertex.

$GPYC/GPYT$ can be seen as the derivative indicators of eigenvector centralities, with their topological meaning similar to the latter (Zhang and Zhang, 2022). The centrality or importance of a certain vertex not only depends on the property of itself, but relies on the properties of its neighbors. Corresponding, within the framework of this article, complex properties (or the knowledge complexity) of the country vertex depends on both the knowledge complexity of other country vertexes (be unlinked each other) and the knowledge complexity of technology vertexes (be linked directly); the same as for a certain technology. The higher the $GPYC/GPYT$ is, the knowledge complexity of a country or technology will be.

Appendix B GPYT and GPYC results

The full results of GPYTs for 35 technology domains and the GPYCs for 100 countries (or regions) in 2021 are shown in Table S.1 and Table S.2 respectively.

Table S.1: Full GPYT results of technology domains in 2021

GPYT Ranking	Technology Domain	GPYT	Patents	Patents Ranking
1	Semiconductors	1.0000	8779	10
2	Audio-visual technology	0.9782	9636	8
3	Digital communication	0.9759	23939	2
4	Optics	0.9551	7425	11
5	Basic communication processes	0.9285	1639	34
6	Telecommunications	0.9064	6186	15
7	Computer technology	0.8753	27212	1
8	Micro-structural and nano-technology	0.8581	354	35
9	Measurement	0.8457	12426	6
10	Macromolecular chemistry, polymers	0.8329	4889	23
11	Surface technology, coating	0.8237	4044	28
12	Electrical machinery, apparatus, energy	0.8161	19261	4
13	Textile and paper machines	0.8055	2423	32
14	Machine tools	0.7827	4297	26
15	Chemical engineering	0.7532	5093	20
16	Control	0.7416	4967	22
17	Mechanical elements	0.7133	4970	21
18	Thermal processes and apparatus	0.7045	3935	29
19	Handling	0.6973	6472	14
20	Transport	0.6953	9883	7
21	Materials, metallurgy	0.6847	4503	25
22	Organic fine chemistry	0.6821	6093	16
23	Furniture, games	0.6791	4732	24
24	Basic materials chemistry	0.6590	5486	18
25	Other special machines	0.6554	6745	12
26	Engines, pumps, turbines	0.6302	4281	27
27	Biotechnology	0.6195	9348	9
28	Civil engineering	0.6174	5801	17
29	Food chemistry	0.6129	2454	31
30	Analysis of biological materials	0.6099	2066	33
31	Other consumer goods	0.6060	6474	13
32	Pharmaceuticals	0.5821	13870	5
33	IT methods for management	0.4828	5320	19
34	Medical technology	0.4335	19414	3
35	Environmental technology	0.0000	2666	30

Table S.2: Full GPYC results of 100 countries in 2021.

GPYC Ranking	Country or Region	GPYC	Patents	Patents Ranking
1	Japan	1.00000	49439.2	3
2	China	0.99778	63202.2	1
3	Singapore	0.93718	1713.8	17
4	Taiwan, PRC	0.91739	406.6	29
5	Netherlands	0.90656	3920.9	10
6	Malaysia	0.86690	136.2	42

GPYC Ranking	Country or Region	GPYC	Patents	Patents Ranking
7	Korea	0.85554	20195.6	4
8	Finland	0.82384	2127.7	13
9	United States	0.80559	57722.6	2
10	Israel	0.78009	2076.6	14
11	Germany	0.71820	16899.2	5
12	Pakistan	0.70634	0.5	96
13	Hong Kong, PRC	0.69830	817.5	25
14	Sweden	0.69239	4107.8	9
15	Ireland	0.69223	919.0	23
16	Belgium	0.67971	1351.1	22
17	Belarus	0.67661	12.3	71
18	France	0.67517	7335.6	6
19	United Kingdom	0.66544	5859.3	7
20	Canada	0.65768	2566.0	12
21	Liechtenstein	0.65362	234.0	37
22	Austria	0.63344	1551.6	19
23	Switzerland	0.61863	5299.8	8
24	India	0.60700	1985.1	15
25	Estonia	0.60416	47.3	51
26	Lebanon	0.60363	5.8	81
27	Russia	0.57890	899.8	24
28	Saudi Arabia	0.57138	789.1	26
29	Cayman Islands	0.56947	275.1	35
30	Thailand	0.56451	147.7	41
31	Romania	0.54873	27.5	65
32	Portugal	0.53892	254.4	36
33	Puerto Rico	0.53576	19.7	67
34	Italy	0.53498	3489.0	11
35	Czechia	0.52963	277.9	33
36	Hungary	0.52904	102.7	45
37	Malta	0.52899	45.5	54
38	Philippines	0.52648	28.9	63
39	Jordan	0.52569	23.0	66
40	Poland	0.52139	353.1	31
41	Latvia	0.52042	38.0	57
42	Armenia	0.51704	4.5	84
43	Lithuania	0.51599	45.5	53
44	Korea, DPR	0.50748	1.0	94
45	Spain	0.50015	1480.9	21
46	Iran	0.49965	275.2	34
47	Luxembourg	0.49810	346.8	32
48	Denmark	0.48913	1522.1	20
49	Ukraine	0.48755	121.9	44
50	Australia	0.48694	1729.7	16
51	Cyprus	0.48633	54.7	49
52	Bulgaria	0.48577	43.8	55

GPYC Ranking	Country or Region	GPYC	Patents	Patents Ranking
53	United Arab Emirates	0.48370	123.8	43
54	Greece	0.47475	83.8	48
55	Tunisia	0.47429	11.5	75
56	Slovakia	0.47328	38.9	56
57	Uruguay	0.46713	5.5	82
58	Iceland	0.46359	30.5	62
59	Egypt	0.45616	46.1	52
60	Sri Lanka	0.45529	38.0	58
61	Brazil	0.45514	605.6	28
62	Venezuela	0.45251	0.0	98
63	Morocco	0.45132	51.9	50
64	Norway	0.44867	706.9	27
65	Slovenia	0.44598	92.5	46
66	Chile	0.44235	156.5	40
67	Argentina	0.44135	28.3	64
68	Georgia	0.43906	15.0	70
69	Monaco	0.43607	5.0	83
70	New Zealand	0.43606	361.1	30
71	Trinidad and Tobago	0.42994	2.5	92
72	Kuwait	0.42347	3.5	88
73	Indonesia	0.42289	7.7	76
74	Jamaica	0.42020	7.5	77
75	Mexico	0.41573	157.9	39
76	Cuba	0.41364	15.5	69
77	Bosnia and Herzegovina	0.41142	12.0	73
78	South Africa	0.39692	221.3	38
79	Croatia	0.39402	30.5	61
80	Uzbekistan	0.38875	3.3	89
81	Costa Rica	0.38818	4.0	85
82	Algeria	0.37928	7.0	78
83	Peru	0.37659	30.8	60
84	Colombia	0.37177	86.0	47
85	Panama	0.36903	12.0	72
86	Turkey	0.36032	1657.9	18
87	Moldova	0.35562	3.5	87
88	Kazakhstan	0.34510	31.0	59
89	North Macedonia	0.34161	6.0	79
90	El Salvador	0.33036	0.0	98
91	Kenya	0.30179	6.0	80
92	Andorra	0.29719	3.2	90
93	Nigeria	0.29534	4.0	85
94	Bermuda	0.29058	19.5	68
95	Guatemala	0.28748	2.0	93
96	Zimbabwe	0.28649	0.3	97
97	Mongolia	0.27585	1.0	94
98	Seychelles	0.25433	3.0	91

GPYC Ranking	Country or Region	GPYC	Patents	Patents Ranking
99	Ecuador	0.22979	11.8	74
100	Djibouti	0.00000	0.0	98

Appendix C Baseline Regression Results and Robustness Checks

C.1 Research Design

Table S.3: Abbreviations of variables and corresponding names and definitions

Type	Variable abbr.	Variable name	Variable definition
Explained Variable	GDPpc_growth	GDP per capita growth	Annual growth rate of GDP per capita
	GNIpc_growth	GNI per capita growth	Annual growth rate of GNI per capita
	consumption	Household consumption	Proportion of household consumption to GDP
	co2	CO_2 emission based on consumption	Logarithmic value of consumption-based CO_2 emission
	CWpc	Comprehensive Wealth per capita	Annual growth rate of Comprehensive Wealth per capita
	HCpc	Human Capital per capita	Annual growth rate of Human Capital per capita
	light	Nighttime light	Aggregated luminance of nighttime light within a specific region
	electricity	electricity production	Annual growth rate of electricity production
	rail	rail lines	Annual growth rate of total rail lines length within a specific region
Core Explanatory Variable	GPYC_rank	GPYC ranking	Ranking percentile of GPYC
Instrumental Variable	LO	Learning Opportunity	Geographic distance with US multiplied by educational expenditure per student in the US
	LP	Learning Probability	Mainstream language (dummy) multiplied by articles published in core journals by the US
Control Variable	government	Government intervention	Proportion of government final consumption expenditure to GDP
	resource	Natural resource	Proportion of the export trade of natural resources to total export trade

Type	Variable abbr.	Variable name	Variable definition
	capital	Capital formation	Proportion of total capital formation to GDP
	industry	Industry production	Proportion of industrial value-added to GDP
	urbanize	Urbanization	Growth rate of urban population
	EI	Employment in industry	Proportion of employment in industry to total employment
	open	Openness of market	Proportion of total trade value of commodities to GDP
	employment	Employment rate	Annual employment rate
	FDI	Foreign Direct Investment	Proportion of net inflows of FDI to GDP
	reserve	International reserve	Proportion of total reserves to GDP
	tiva	Trade with China	Proportion of bilateral trade value with China to GDP
	coop	Technology cooperation with China	Logarithmic number of granted patents cooperated with China

Data source: World Bank Open Data; World Economic Outlook; World Development Indicators (WDI); Global Carbon Budget; UN Comtrade; Organization for Economic Co-operation and Development (OECD) statistics; OECD Inter-Country Input-Output table; World Intellectual Property Organization (WIPO) PATENTSCOPE; Inter-Country Risk Guidance (ICRG); Penn World Table; International Military Fund (IMF) International Financial Statistics; International Labour Organization (ILO); DMSP/OLS; NPP-VIIRS; Global Electrification database; International Union of Railways (UIC) Railisa Database; International Civil Aviation Organization (ICAO); the authors' calculations.

Table S.4: Abbreviations of variables and corresponding names and definitions

Type	Variable abbr.	Variable name	Variable definition
Explanatory Variable	GPYC	GPYC value	Max-Min Standardized Value of GPYC
	GPYCr_15y	GPYC ranking (15 Years)	Ranking percentile of GPYC generated by patents within 15 years
	GPYCr_10y	GPYC ranking (10 Years)	Ranking percentile of GPYC generated by patents within 10 years
	GPYCr_5y	GPYC ranking (5 Years)	Ranking percentile of GPYC generated by patents within 5 years
Moderating Variable	BRI	Belt and Road	If a country join the "Belt and Road" Initiative in a specific year
	ideal	Ideal point distance	Average Ideal point distance with other countries
	geopolit	Geo-political risk	External total geo-political risk a country confront with
	wtar	Tariff	Weighted average tariff a country confront with
Control Variable	TFP	Total Factor Productivity	Total Factor Productivity adjusted for inflation

Type	Variable abbr.	Variable name	Variable definition
	education	Education expenditure	Proportion of Expenditure on education to GDP
	enroll	School enrollment	School enrollment rate of tertiary education
	gov.eff	Government effectiveness	Point estimates of perceptions of the quality of public services
	law.order	Law and order situation	Assessment of law and order situation
	acct	Government accountability	Assessment of government accountability
	technicians	Technicians in R&D	Technicians in R&D per million people
	R.D_expenditure	Expenditure on R&D	Proportion of Expenditure on R&D to GDP
	charge	Charge for Intellectual Property	Proportion of charge for Intellectual Property to GDP
	optimize	Industrial optimization	Proportion of value added of secondary and tertiary industries to GDP

Data source: Penn World Table; World Bank Open Data; World Governance Indicators (WGI); UNESCO Institute of Statistics (UIS); IMF Balance of Payments Statistics Yearbook; Inter-Country Risk Guidance (ICRG); United Nations General Assembly voting data; Caldara and Iacoviello (2022); CEPII The Trade and Production Database (TradeProd); the authors' calculations.

Table S.5: Descriptive statistics of each variable for robustness checks

Statistic	N	Mean	St. Dev.	Min	Max
GPYC_01	1,312	0.078	0.042	0.000	0.624
TFP	970	0.938	0.215	0.495	1.782
education	942	4.243	1.347	0.964	10.250
enroll	1,088	37.780	15.210	2.010	52.490
gov.eff.	943	0.316	0.544	−0.120	2.234
law.order	1,280	3.508	1.572	0.000	6.000
acct	1,280	3.517	1.979	0.000	6.000
technicians	492	2.400	0.489	1.017	3.402
R.D_expenditure	856	0.814	0.790	0.016	5.706
charge	800	7.442	1.033	2.301	10.100
optimize	1,093	2.233	0.170	1.712	2.897
GPYCr_15y	1,312	0.491	0.259	0.010	1.000
GPYCr_10y	1,312	0.492	0.258	0.010	1.000
GPYCr_5y	1,312	0.494	0.258	0.010	1.000

Table S.6: Descriptive statistics of each variable for Global sample and moderating effect model.

Statistic	N	Mean	St. Dev.	Min	Max
GDPpc_growth	3,168	0.020	0.042	−0.127	0.149
GNIpc_growth	3,168	0.022	0.039	−0.121	0.148
household	2,752	0.033	0.071	−0.127	0.432
CO2	2,636	1.689	0.959	−0.127	3.465
DCWpc	2,115	0.046	0.114	−0.127	0.489
HCpc	2,125	0.056	0.120	−0.127	0.504
light	2,763	0.097	0.424	−0.127	4.728
elec	2,912	7.975	1.108	4.770	10.424
rail	1,587	8.394	1.362	5.613	12.178
BRI	3,200	0.410	0.492	0	1
wtar	2,774	0.021	0.020	0.002	0.113
geopolit	2,880	0.700	0.299	0.000	1.000
ideal	2,433	0.555	0.744	−0.127	2.782
GPYC_rank	3,200	0.505	0.289	0.010	1.000
LP	2,156	0.066	0.102	0.000	0.286
LO	3,136	14.553	0.715	10.859	14.932
government	2,848	0.771	0.122	0.430	1.171
resource	2,614	0.035	0.048	0.0003	0.310
capital	2,850	0.239	0.068	0.053	0.466
industry	2,829	0.277	0.094	0.064	0.598
urbanize	3,168	0.013	0.015	−0.030	0.072
EI	2,883	0.236	0.067	0.048	0.420
open	2,909	0.666	0.466	0.131	3.308
employment	2,883	0.547	0.102	0.245	0.800
FDI	2,876	0.098	0.505	−0.106	5.549
reserve	2,729	0.101	0.124	0.004	0.930
tiva	1,714	0.014	0.017	0.000	0.128
coop	1,260	2.659	2.466	0.000	9.508

Table S.7: Result of first stage of baseline regressions

Variable	Coef.	Std. Error	t value	p value
LP	0.146*	0.078	1.873	[0.062]
LO	-0.125***	0.012	-10.590	[0.000]
government	0.234***	0.079	2.964	[0.003]
resource	1.338***	0.120	11.111	[0.000]
capital	0.214**	0.086	2.498	[0.013]
industry	-0.244***	0.083	-2.937	[0.003]
urbanize	-1.620***	0.402	-4.026	[0.000]
EI	-0.590***	0.094	-6.305	[0.000]
open	-0.075***	0.010	-7.646	[0.000]
employment	0.024	0.073	0.332	[0.740]
FDI	0.152***	0.025	6.188	[0.000]
reserve	-0.113***	0.043	-2.637	[0.009]
tiva	-3.749***	0.584	-6.425	[0.000]
coop	0.035***	0.002	16.639	[0.000]
Observation		596		
Year FE		596		
Country FE		596		
Adj. R^2		0.855		
Cragg-Donald Wald F		100.7***		
Kleibergen-Paap Wald F		2.965***		

*, **, *** denote p<0.1, <0.05, <0.01.

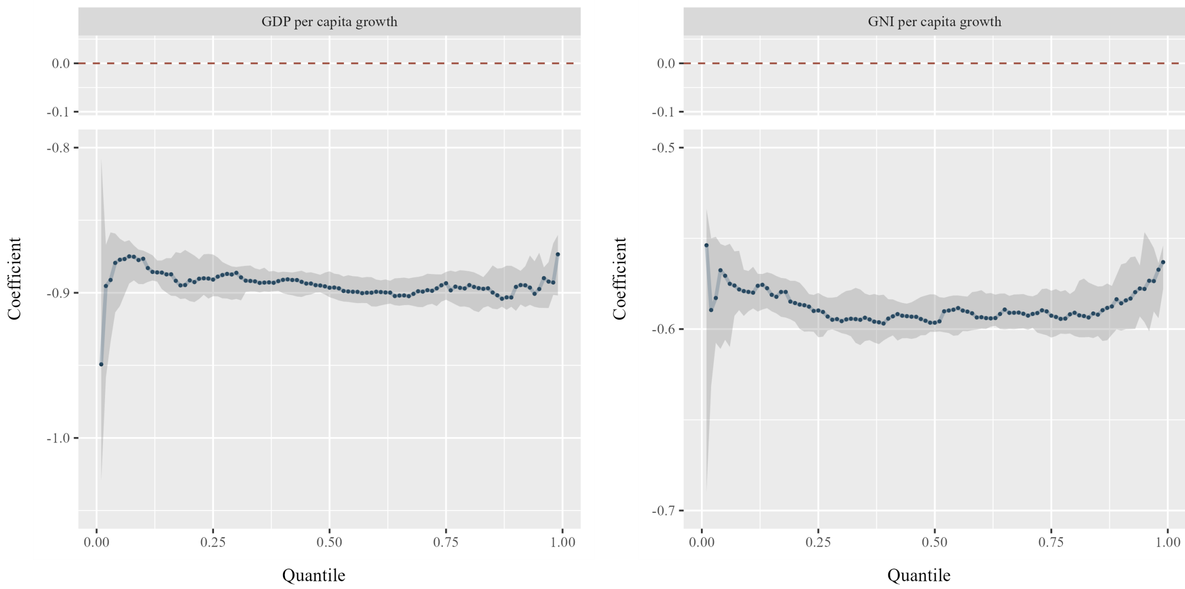


Figure S.1: Full results of baseline regressions

Table S.8: Results of baseline regressions with GDPpc_growth as explained variable.

Variable	GDPpc_growth						
	(1) tau=0.2	(2) tau=0.3	(3) tau=0.4	(4) tau=0.5	(5) tau=0.6	(6) tau=0.7	(7) tau=0.8
GPYC_rank	−0.891***	−0.886***	−0.891***	−0.897***	−0.899***	−0.899***	−0.895***
	(0.008)	(0.006)	(0.005)	(0.005)	(0.005)	(0.006)	(0.007)
government	0.274***	0.250***	0.245***	0.247***	0.242***	0.248***	0.261***
	(0.018)	(0.019)	(0.018)	(0.021)	(0.021)	(0.019)	(0.023)
resource	1.326***	1.366***	1.351***	1.360***	1.328***	1.341***	1.404***
	(0.053)	(0.041)	(0.025)	(0.035)	(0.038)	(0.045)	(0.048)
capital	0.527***	0.504***	0.506***	0.493***	0.492***	0.503***	0.521***
	(0.022)	(0.022)	(0.020)	(0.023)	(0.023)	(0.025)	(0.036)
industry	0.036*	0.021	0.025*	0.021	0.025	0.038**	0.055**
	(0.021)	(0.018)	(0.015)	(0.018)	(0.019)	(0.018)	(0.022)
urbanize	−3.428***	−3.347***	−3.373***	−3.311***	−3.349***	−3.298***	−3.375***
	(0.102)	(0.095)	(0.079)	(0.102)	(0.101)	(0.088)	(0.090)
EI	−0.508***	−0.508***	−0.524***	−0.498***	−0.520***	−0.536***	−0.545***
	(0.032)	(0.026)	(0.021)	(0.025)	(0.026)	(0.028)	(0.033)
open	0.002	0.004	0.006**	0.005	0.007**	0.008***	0.007*
	(0.004)	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)	(0.004)
employment	0.001	−0.006	−0.021	−0.023	−0.030	−0.031	−0.027
	(0.024)	(0.020)	(0.017)	(0.020)	(0.020)	(0.020)	(0.025)
FDI	0.122***	0.118***	0.117***	0.124***	0.124***	0.123***	0.126***
	(0.002)	(0.006)	(0.007)	(0.009)	(0.003)	(0.005)	(0.002)
reserve	−0.112***	−0.127***	−0.137***	−0.140***	−0.147***	−0.143***	−0.136***
	(0.015)	(0.012)	(0.011)	(0.013)	(0.013)	(0.014)	(0.017)
tiva	0.000	0.000	0.000	0.000	0.000*	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
coop	0.033***	0.033***	0.033***	0.032***	0.032***	0.032***	0.033***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Observation	596	596	596	596	596	596	596
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.901	0.902	0.903	0.902	0.901	0.901	0.901

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.9: Results of baseline regressions with GNIpc_growth as explained variable.

Variable	GNIpc_growth						
	(1) tau=0.2	(2) tau=0.3	(3) tau=0.4	(4) tau=0.5	(5) tau=0.6	(6) tau=0.7	(7) tau=0.8
GPYC_rank	−0.586***	−0.596***	−0.594***	−0.596***	−0.593***	−0.593***	−0.591***
	(0.005)	(0.005)	(0.006)	(0.007)	(0.006)	(0.006)	(0.008)
government	0.114***	0.126***	0.130***	0.127***	0.130***	0.139***	0.143***
	(0.025)	(0.016)	(0.021)	(0.022)	(0.020)	(0.023)	(0.024)
resource	0.916***	0.934***	0.979***	0.986***	0.997***	0.964***	1.029***
	(0.038)	(0.039)	(0.043)	(0.045)	(0.040)	(0.046)	(0.058)
capital	0.359***	0.373***	0.380***	0.398***	0.410***	0.412***	0.448***
	(0.029)	(0.024)	(0.026)	(0.027)	(0.022)	(0.026)	(0.030)
industry	−0.072***	−0.053***	−0.059***	−0.041**	−0.039**	−0.026	−0.013
	(0.020)	(0.018)	(0.018)	(0.020)	(0.019)	(0.020)	(0.023)
urbanize	−1.703***	−1.794***	−1.863***	−1.841***	−1.903***	−2.049***	−2.037***
	(0.094)	(0.069)	(0.089)	(0.101)	(0.094)	(0.103)	(0.116)
EI	−0.364***	−0.356***	−0.345***	−0.337***	−0.362***	−0.388***	−0.396***
	(0.030)	(0.028)	(0.027)	(0.029)	(0.030)	(0.030)	(0.031)
open	0.013***	0.011***	0.010***	0.010***	0.010***	0.013***	0.011***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
employment	0.016	0.022	0.033*	0.003	0.021	0.015	0.020
	(0.016)	(0.017)	(0.019)	(0.021)	(0.020)	(0.019)	(0.022)
FDI	0.063***	0.066***	0.069***	0.077***	0.079***	0.091***	0.089***
	(0.006)	(0.011)	(0.011)	(0.010)	(0.017)	(0.013)	(0.002)
reserve	−0.085***	−0.107***	−0.118***	−0.123***	−0.126***	−0.117***	−0.133***
	(0.015)	(0.014)	(0.013)	(0.014)	(0.013)	(0.015)	(0.017)
tiva	0.000***	0.000***	0.000***	0.000**	0.000*	0.000***	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
coop	0.021***	0.023***	0.023***	0.023***	0.023***	0.023***	0.024***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Observation	596	596	596	596	596	596	596
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.843	0.843	0.845	0.845	0.844	0.846	0.848

*, **, *** denote p<0.1, <0.05, <0.01.

C.2 Results of Baseline Regressions

C.3 Robustness Checks Design

Table S.10: Timelines of countries joining the Belt and Road Initiative.

Year	Country
2013	Belarus, Cambodia, China, PRC, Kyrgyz Republic, Moldova, Mongolia, North Macedonia, Pakistan
2014	Thailand
2015	Armenia, Azerbaijan, Bulgaria, Cameroon, Comoros, Czech Republic, Hungary, Indonesia, Iraq, Kazakhstan, Poland, Romania, Russian Federation, Serbia, Slovak Republic, Somalia, South Africa, Turkey, Uzbekistan
2016	Egypt, Arab Rep., Georgia, Latvia, Myanmar, Papua New Guinea
2017	Albania, Bosnia and Herzegovina, Côte d'Ivoire, Croatia, Estonia, Kenya, Lebanon, Lithuania, Madagascar, Malaysia, Maldives, Montenegro, Morocco, Nepal, New Zealand, Panama, Philippines, Slovenia, Sri Lanka, Timor-Leste, Turkmenistan
2018	Algeria, Angola, Antigua and Barbuda, Bahrain, Benin, Bolivia, Brunei Darussalam, Burundi, Cabo Verde, Chad, Chile, Congo, Rep., Cook Islands, Costa Rica, Djibouti, Dominica, Ecuador, El Salvador, Ethiopia, Fiji, Gabon
2019	Austria, Bangladesh, Barbados, Cuba, Cyprus, Dominican Republic, Equatorial Guinea, Italy, Jamaica, Lesotho, Liberia, Luxembourg, Mali, Peru, Qatar, Solomon Islands
2020	Kiribati
2021	Botswana, Central African Republic, Congo, Dem. Rep., Eritrea, Guinea-Bissau
2022	Argentina, Malawi, Nicaragua, Syrian Arab Republic
2023	Afghanistan, Honduras, Jordan
2024	Nauru
2025	Colombia

Table S.11: All Prospective Founding Members of AIIB.

Country	ISO code	Intention	Signing Date
Country	ISO code	Intention	Signature
Azerbaijan	AZE	15-Apr-15	29-Jun-15
Iceland	ISL	15-Apr-15	29-Jun-15
Israel	ISR	15-Apr-15	29-Jun-15
Poland	POL	15-Apr-15	9-Oct-15
Portugal	PRT	15-Apr-15	29-Jun-15
South Africa	ZAF	15-Apr-15	3-Dec-15
Sweden	SWE	15-Apr-15	29-Jun-15
Egypt	EGY	14-Apr-15	29-Jun-15
Norway	NOR	14-Apr-15	29-Jun-15

Country	ISO code	Intention	Signing Date
Russia	RUS	14-Apr-15	29-Jun-15
Brazil	BRA	12-Apr-15	29-Jun-15
Denmark	DNK	12-Apr-15	27-Oct-15
Finland	FIN	12-Apr-15	29-Jun-15
Georgia	GEO	12-Apr-15	29-Jun-15
Netherlands	NLD	12-Apr-15	29-Jun-15
Austria	AUT	11-Apr-15	29-Jun-15
South Korea	KOR	11-Apr-15	29-Jun-15
Spain	ESP	11-Apr-15	29-Jun-15
Turkey	TUR	10-Apr-15	29-Jun-15
Kyrgyzstan	KGZ	9-Apr-15	29-Jun-15
Malta	MLT	9-Apr-15	29-Jun-15
Iran	IRN	7-Apr-15	29-Jun-15
United Arab Emirates	ARE	5-Apr-15	29-Jun-15
Australia	AUS	3-Apr-15	29-Jun-15
France	FRA	2-Apr-15	29-Jun-15
Italy	ITA	2-Apr-15	29-Jun-15
Germany	DEU	1-Apr-15	29-Jun-15
Switzerland	CHE	28-Mar-15	29-Jun-15
United Kingdom	GBR	28-Mar-15	29-Jun-15
Luxembourg	LUX	27-Mar-15	29-Jun-15
Jordan	JOR	7-Feb-15	29-Jun-15
Saudi Arabia	SAU	13-Jan-15	29-Jun-15
Tajikistan	TJK	13-Jan-15	29-Jun-15
New Zealand	NZL	5-Jan-15	29-Jun-15
Maldives	MDV	31-Dec-14	29-Jun-15
Indonesia	IDN	25-Nov-14	29-Jun-15
Bangladesh	BGD	24-Oct-14	29-Jun-15
Brunei	BRN	24-Oct-14	29-Jun-15
Cambodia	KHM	24-Oct-14	29-Jun-15
China	CHN	24-Oct-14	29-Jun-15
India	IND	24-Oct-14	29-Jun-15
Kazakhstan	KAZ	24-Oct-14	29-Jun-15
Kuwait	KWT	24-Oct-14	4-Dec-15
Laos	LAO	24-Oct-14	29-Jun-15
Malaysia	MYS	24-Oct-14	21-Aug-15
Mongolia	MNG	24-Oct-14	29-Jun-15
Myanmar	MMR	24-Oct-14	29-Jun-15
Nepal	NPL	24-Oct-14	29-Jun-15
Oman	OMN	24-Oct-14	29-Jun-15
Pakistan	PAK	24-Oct-14	29-Jun-15
Philippines	PHL	24-Oct-14	31-Dec-15
Qatar	QAT	24-Oct-14	29-Jun-15
Singapore	SGP	24-Oct-14	29-Jun-15
Sri Lanka	LKA	24-Oct-14	29-Jun-15
Thailand	THA	24-Oct-14	1-Oct-15

Country	ISO code	Intention	Signing Date
Uzbekistan	UZB	24-Oct-14	29-Jun-15
Vietnam	VNM	24-Oct-14	29-Jun-15

Table S.12: Countries of Bilateral Community of Common Destiny.

Bilateral Community of Common Destiny	Country	ISO code
New Era All-Weather Community of Shared Future	Cambodia	KHM
High-quality, High-level, High-standard New Era Community of Shared Future	Laos	LAO
Closer Community of Shared Future in the New Era	Pakistan	PAK
New Era Community of Shared Future	Egypt	EGY
New Era Community of Shared Future	Serbia	SRB
New Era Community of Shared Future	Solomon Islands	SLB
New Era Community of Shared Future	Kenya	KEN
High-level Community of Shared Future	South Africa	ZAF
High-level Community of Shared Future	Republic of the Congo	COG
High-level Community of Shared Future	Zimbabwe	ZWE
High-level Community of Shared Future	Senegal	SEN
High-level Community of Shared Future	Nigeria	NGA
Community of Shared Future	Cuba	CUB
Community of Shared Future	Uzbekistan	UZB
Community of Shared Future	Turkmenistan	TKM
Community of Shared Future	Myanmar	MMR
Community of Shared Future	Malaysia	MYS
Generations of Friendship, High Trust, Shared Destiny Community of Shared Future	Kazakhstan	KAZ
Generations of Friendship, Shared Destiny, Mutual Benefit Community of Shared Future	Tajikistan	TJK
Neighborly Friendship, Shared Prosperity Community of Shared Future	Kyrgyzstan	KGZ
Peaceful Coexistence, Mutual Support, Win-win Cooperation Community of Shared Future	Mongolia	MNG
Community of Shared Future for a Fairer World and More Sustainable Planet	Brazil	BRA
More Stable, More Prosperous, More Sustainable Community of Shared Future	Thailand	THA
Strategic Community of Shared Future	Vietnam	VNM
Regionally and Globally Influential Community of Shared Future	Indonesia	IDN

Table S.13: Countries and their bilateral diplomatic relationship with China.

Bilateral diplomatic relationship	ISO code	Country
New Era All-weather Comprehensive Strategic Partnership	HUN	Hungary
	UZB	Uzbekistan
All-weather Comprehensive Strategic Partnership	BLR	Belarus
All-weather Strategic Cooperative Partnership	PAK	Pakistan
All-weather Strategic Partnership	VEN	Venezuela
	ETH	Ethiopia
Permanent Comprehensive Strategic Partnership	KAZ	Kazakhstan
New Era All-round Strategic Cooperative Partnership	ZAF	South Africa
New Era Comprehensive Strategic Cooperative Partnership	TJK	Tajikistan
Comprehensive Strategic Cooperative Partnership	KHM	Cambodia
	VNM	Vietnam
	MMR	Myanmar
	THA	Thailand
	SEN	Senegal
	NAM	Namibia
	MOZ	Mozambique
	SLE	Sierra Leone
	ZWE	Zimbabwe
	KEN	Kenya
	TZA	Tanzania
	GAB	Gabon
	COD	DR Congo
	COG	Congo Republic
	ZMB	Zambia
	MDV	Maldives
	BGD	Bangladesh
	GIN	Guinea
	GNQ	Equatorial Guinea
	AGO	Angola
	MDG	Madagascar
New Era Comprehensive Strategic Coordination Partnership	RUS	Russia
New Era Comprehensive Strategic Partnership	KGZ	Kyrgyzstan
Comprehensive Strategic Partnership	MYS	Malaysia
	MNG	Mongolia

Bilateral diplomatic relationship	ISO code	Country
	IDN	Indonesia
	DZA	Algeria
	EGY	Egypt
	DNK	Denmark
	FRA	France
	PRT	Portugal
	ESP	Spain
	GRC	Greece
	ITA	Italy
	MEX	Mexico
	ARG	Argentina
	BRA	Brazil
	PER	Peru
	AUS	Australia
	NZL	New Zealand
	SAU	Saudi Arabia
	IRN	Iran
	SRB	Serbia
	POL	Poland
	ECU	Ecuador
	ARE	United Arab Emirates
	CHL	Chile
	TKM	Turkmenistan
	BHR	Bahrain
	DJI	Djibouti
	NGA	Nigeria
	TLS	Timor-Leste
	TGO	Togo
	RWA	Rwanda
	URY	Uruguay
Global Comprehensive Strategic Partnership Oriented toward the 21st Century	GBR	United Kingdom
All-round Strategic Partnership	DEU	Germany

C.4 Results of Robustness Checks

Table S.14: Robustness checks results based on household consumption.

Variable	Economic Growth												
	Core X Replacement	Anticipation Effect	Time-lag Effect			Additional Control Variable		Period Adjustment		Country Adjustment			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel A: tau=0.25													
GPYC_rank	8.590***	0.000	-0.251***	-0.198***	-0.145***	-0.252***	-0.200***	-0.297***	-0.710***	-0.327***	-0.112***	-0.327***	-0.074***
	(0.186)	(0.019)	(0.019)	(0.019)	(0.018)	(0.025)	(0.024)	(0.023)	(0.025)	(0.022)	(0.023)	(0.022)	(0.024)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	737	594	567	538	509	466	445	478	447	516	356	516	416
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel B: tau=0.50													
GPYC_rank	9.022***	0.000	-0.263***	-0.223***	-0.147***	-0.266***	-0.215***	-0.289***	-0.689***	-0.320***	-0.111***	-0.320***	-0.096***
	(0.205)	(0.021)	(0.021)	(0.020)	(0.020)	(0.028)	(0.028)	(0.027)	(0.030)	(0.027)	(0.028)	(0.027)	(0.028)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	737	594	567	538	509	466	445	478	447	516	356	516	416
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel C: tau=0.75													
GPYC_rank	8.861***	-0.001	-0.258***	-0.190***	-0.134***	-0.250***	-0.177***	-0.287***	-0.587***	-0.295***	-0.105***	-0.295***	-0.060**
	(0.186)	(0.021)	(0.021)	(0.020)	(0.020)	(0.027)	(0.028)	(0.027)	(0.030)	(0.027)	(0.028)	(0.027)	(0.030)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	737	594	567	538	509	466	445	478	447	516	356	516	416
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.15: Robustness checks results based on CO_2 emission calculated via consumption.

Variable	Economic Growth												
	Core X Replacement	Anticipation Effect	Time-lag Effect			Additional Control Variable		Period Adjustment		Country Adjustment			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel A: tau=0.25													
GPYC_rank	-21.163***	-0.290***	-2.192***	-2.858***	-3.530***	-2.737***	-3.054***	-1.548***	-1.628***	-1.888***	-2.427***	-1.888***	0.000
	(0.204)	(0.029)	(0.035)	(0.036)	(0.037)	(0.037)	(0.038)	(0.035)	(0.036)	(0.036)	(0.038)	(0.036)	(0.036)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	748	595	596	596	567	466	445	479	447	516	356	516	417
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel B: tau=0.50													
GPYC_rank	-20.798***	-0.373***	-2.322***	-2.897***	-3.594***	-2.741***	-2.929***	-1.642***	-1.840***	-1.933***	-2.189***	-1.933***	-0.148***
	(0.232)	(0.031)	(0.039)	(0.041)	(0.043)	(0.044)	(0.045)	(0.041)	(0.040)	(0.043)	(0.047)	(0.043)	(0.039)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	748	595	596	596	567	466	445	479	447	516	356	516	417
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel C: tau=0.75													
GPYC_rank	-20.938***	-0.371***	-2.335***	-2.931***	-3.579***	-2.712***	-2.939***	-1.458***	-1.734***	-1.926***	-1.906***	-1.926***	-0.048
	(0.200)	(0.031)	(0.039)	(0.040)	(0.042)	(0.041)	(0.041)	(0.041)	(0.040)	(0.040)	(0.042)	(0.040)	(0.039)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	748	595	596	596	567	466	445	479	447	516	356	516	417
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.16: Robustness checks results based on Comprehensive Wealth per capita.

Variable	Economic Growth												
	Core X Replacement	Anticipation Effect	Time-lag Effect			Additional Control Variable		Period Adjustment		Country Adjustment			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel A: tau=0.25													
GPYC_rank	21.147***	0.000	-0.779***	-0.583***	-0.271***	-0.218***	-0.168***	-0.496***	-0.052	-0.423***	-0.522***	-0.423***	-0.271***
	(0.201)	(0.031)	(0.028)	(0.029)	(0.030)	(0.032)	(0.033)	(0.032)	(0.032)	(0.031)	(0.034)	(0.031)	(0.033)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	754	573	576	548	520	446	426	463	432	496	336	496	397
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel B: tau=0.50													
GPYC_rank	21.572***	0.000	-0.783***	-0.567***	-0.250***	-0.241***	-0.171***	-0.530***	-0.089***	-0.408***	-0.566***	-0.408***	-0.303***
	(0.226)	(0.031)	(0.031)	(0.033)	(0.033)	(0.035)	(0.036)	(0.035)	(0.034)	(0.033)	(0.034)	(0.033)	(0.036)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	754	573	576	548	520	446	426	463	432	496	336	496	397
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel C: tau=0.75													
GPYC_rank	20.962***	0.000	-0.714***	-0.501***	-0.198***	-0.209***	-0.132***	-0.551***	-0.068**	-0.376***	-0.543***	-0.376***	-0.310***
	(0.200)	(0.028)	(0.032)	(0.033)	(0.032)	(0.031)	(0.034)	(0.033)	(0.030)	(0.030)	(0.030)	(0.030)	(0.033)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	754	573	576	548	520	446	426	463	432	496	336	496	397
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.17: Robustness checks results based on Human Capital per capita.

Variable	Economic Growth												
	Core X Replacement	Anticipation Effect	Time-lag Effect			Additional Control Variable		Period Adjustment		Country Adjustment			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel A: tau=0.25													
GPYC_rank	24.382***	0.000	-0.803***	-0.561***	-0.274***	-0.367***	-0.251***	-0.500***	-0.210***	-0.386***	-0.531***	-0.386***	-0.212***
	(0.201)	(0.030)	(0.028)	(0.029)	(0.030)	(0.032)	(0.034)	(0.033)	(0.031)	(0.031)	(0.034)	(0.031)	(0.034)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	754	575	576	548	520	446	426	463	432	496	336	496	397
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel B: tau=0.50													
GPYC_rank	25.155***	-0.032	-0.826***	-0.565***	-0.283***	-0.382***	-0.294***	-0.570***	-0.241***	-0.410***	-0.612***	-0.410***	-0.296***
	(0.225)	(0.031)	(0.032)	(0.033)	(0.033)	(0.036)	(0.038)	(0.036)	(0.033)	(0.033)	(0.035)	(0.033)	(0.036)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	754	575	576	548	520	446	426	463	432	496	336	496	397
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel C: tau=0.75													
GPYC_rank	24.454***	-0.039	-0.729***	-0.540***	-0.278***	-0.310***	-0.196***	-0.570***	-0.216***	-0.371***	-0.594***	-0.371***	-0.280***
	(0.201)	(0.027)	(0.031)	(0.032)	(0.032)	(0.034)	(0.036)	(0.034)	(0.031)	(0.031)	(0.030)	(0.031)	(0.033)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	754	575	576	548	520	446	426	463	432	496	336	496	397
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.18: Robustness checks results based on nighttime light.

Variable	Economic Growth												
	Core X Replacement	Anticipation Effect	Time-lag Effect			Additional Control Variable		Period Adjustment		Country Adjustment			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel A: tau=0.25													
GPYC_rank	81.987***	-0.681***	-2.012***	-1.036***	-0.057**	-1.379***	-0.724***	-1.383***	-0.068**	-0.184***	-0.438***	-0.184***	-1.222***
	(0.209)	(0.031)	(0.033)	(0.030)	(0.029)	(0.040)	(0.040)	(0.037)	(0.034)	(0.030)	(0.034)	(0.030)	(0.034)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	753	751	714	676	638	465	443	568	573	634	464	634	523
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel B: tau=0.50													
GPYC_rank	82.079***	-0.664***	-2.015***	-1.011***	-0.045	-1.242***	-0.725***	-1.405***	-0.142***	-0.220***	-0.427***	-0.220***	-1.269***
	(0.236)	(0.033)	(0.034)	(0.033)	(0.033)	(0.045)	(0.045)	(0.040)	(0.038)	(0.035)	(0.038)	(0.035)	(0.039)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	753	751	714	676	638	465	443	568	573	634	464	634	523
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel C: tau=0.75													
GPYC_rank	81.122***	-0.567***	-1.690***	-0.996***	0.000	-1.064***	-0.672***	-1.270***	-0.072**	-0.128***	-0.389***	-0.128***	-1.139***
	(0.207)	(0.030)	(0.035)	(0.032)	(0.032)	(0.039)	(0.039)	(0.035)	(0.035)	(0.033)	(0.036)	(0.033)	(0.037)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	753	751	714	676	638	465	443	568	573	634	464	634	523
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.19: Robustness checks results based on electricity production.

Variable	Economic Growth												
	Core X Replacement	Anticipation Effect	Time-lag Effect			Additional Control Variable		Period Adjustment		Country Adjustment			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel A: tau=0.25													
GPYC_rank	2.215***	-0.234***	-0.904***	-0.913***	-0.930***	-0.748***	-0.694***	-0.821***	-0.622***	-0.182***	-1.288***	-0.182***	-0.065*
	(0.196)	(0.028)	(0.030)	(0.030)	(0.030)	(0.036)	(0.038)	(0.031)	(0.035)	(0.029)	(0.039)	(0.029)	(0.034)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	816	595	596	596	567	466	445	479	447	516	356	516	417
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel B: tau=0.50													
GPYC_rank	2.897***	-0.309***	-0.942***	-0.950***	-0.961***	-0.930***	-0.921***	-0.850***	-0.750***	-0.299***	-1.368***	-0.299***	-0.249***
	(0.222)	(0.028)	(0.032)	(0.032)	(0.032)	(0.042)	(0.043)	(0.035)	(0.037)	(0.032)	(0.046)	(0.032)	(0.035)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	816	595	596	596	567	466	445	479	447	516	356	516	417
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel C: tau=0.75													
GPYC_rank	3.373***	-0.215***	-0.964***	-0.953***	-0.878***	-1.016***	-1.005***	-0.741***	-0.815***	-0.273***	-1.222***	-0.273***	-0.114***
	(0.194)	(0.030)	(0.034)	(0.033)	(0.035)	(0.038)	(0.038)	(0.037)	(0.037)	(0.034)	(0.043)	(0.034)	(0.036)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	816	595	596	596	567	466	445	479	447	516	356	516	417
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.20: Robustness checks results based on rail lines.

Variable	Economic Growth												
	Core X Replacement	Anticipation Effect	Time-lag Effect			Additional Control Variable		Period Adjustment		Country Adjustment			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel A: tau=0.25													
GPYC_rank	13.843***	-1.152***	-5.815***	-6.142***	-6.363***	-9.140***	-8.914***	-5.370***	-5.923***	-6.520***	-8.702***	-6.520***	-9.438***
	(0.204)	(0.022)	(0.036)	(0.038)	(0.039)	(0.041)	(0.041)	(0.039)	(0.036)	(0.038)	(0.042)	(0.038)	(0.041)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	619	623	589	561	530	426	408	446	480	509	380	509	466
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel B: tau=0.50													
GPYC_rank	14.664***	-1.147***	-5.809***	-6.135***	-6.363***	-8.432***	-8.934***	-5.342***	-5.770***	-6.643***	-8.029***	-6.643***	-9.454***
	(0.223)	(0.024)	(0.034)	(0.035)	(0.036)	(0.048)	(0.047)	(0.039)	(0.042)	(0.041)	(0.050)	(0.041)	(0.046)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	619	623	589	561	530	426	408	446	480	509	380	509	466
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Panel C: tau=0.75													
GPYC_rank	14.216***	-1.104***	-5.523***	-5.750***	-5.899***	-7.773***	-8.235***	-4.580***	-5.240***	-6.045***	-6.088***	-6.045***	-7.523***
	(0.206)	(0.027)	(0.036)	(0.038)	(0.039)	(0.044)	(0.044)	(0.041)	(0.041)	(0.042)	(0.043)	(0.042)	(0.043)
Control Variable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observation	619	623	589	561	530	426	408	446	480	509	380	509	466
Year FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

*, **, *** denote p<0.1, <0.05, <0.01.

Appendix D Mechanism Tests and Heterogeneous Analyses

D.1 Results of Mechanism Tests

Table S.21: Results of samples grouped by charges for intellectual property

Dependent Variable	Sample	Quantile	Coef.	Conf. lwr	Conf. upr	p-value
Charge for IP	Total	tau=0.2	-0.469***	-0.566	-0.371	[0.000]
		tau=0.3	-0.510***	-0.593	-0.427	[0.000]
		tau=0.4	-0.434***	-0.516	-0.353	[0.000]
		tau=0.5	-0.451***	-0.529	-0.374	[0.000]
		tau=0.6	-0.387***	-0.452	-0.322	[0.000]
		tau=0.7	-0.363***	-0.441	-0.284	[0.000]
		tau=0.8	-0.288***	-0.388	-0.188	[0.000]
GDPpc_growth	Low charge	tau=0.2	-0.272***	-0.285	-0.259	[0.000]
		tau=0.3	-0.277***	-0.288	-0.266	[0.000]
		tau=0.4	-0.271***	-0.279	-0.263	[0.000]
		tau=0.5	-0.270***	-0.279	-0.261	[0.000]
		tau=0.6	-0.272***	-0.282	-0.263	[0.000]
		tau=0.7	-0.279***	-0.289	-0.268	[0.000]
		tau=0.8	-0.279***	-0.293	-0.266	[0.000]
	High charge	tau=0.2	0.0323***	0.023	0.042	[0.000]
		tau=0.3	0.0394***	0.032	0.046	[0.000]
		tau=0.4	0.0449***	0.038	0.052	[0.000]
		tau=0.5	0.0466***	0.040	0.053	[0.000]
		tau=0.6	0.0480***	0.042	0.054	[0.000]
		tau=0.7	0.0472***	0.040	0.054	[0.000]
		tau=0.8	0.0496***	0.040	0.059	[0.000]

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.22: Results of samples grouped by industrial optimization

Dependent Variable	Sample	Quantile	Coef.	Conf. lwr	Conf. upr	p-value
Industrial optimization	Total	tau=0.2	-0.524***	-0.534	-0.514	[0.000]
		tau=0.3	-0.513***	-0.522	-0.505	[0.000]
		tau=0.4	-0.514***	-0.521	-0.507	[0.000]
		tau=0.5	-0.511***	-0.517	-0.504	[0.000]
		tau=0.6	-0.507***	-0.513	-0.500	[0.000]
		tau=0.7	-0.503***	-0.510	-0.495	[0.000]
		tau=0.8	-0.499***	-0.509	-0.490	[0.000]
GDPpc_growth	Low optimization	tau=0.2	-0.645***	-0.659	-0.631	[0.000]
		tau=0.3	-0.649***	-0.659	-0.639	[0.000]
		tau=0.4	-0.646***	-0.654	-0.638	[0.000]
		tau=0.5	-0.646***	-0.654	-0.638	[0.000]
		tau=0.6	-0.649***	-0.658	-0.640	[0.000]
		tau=0.7	-0.649***	-0.659	-0.640	[0.000]
		tau=0.8	-0.646***	-0.656	-0.637	[0.000]
	High optimization	tau=0.2	0.0801***	0.072	0.088	[0.000]
		tau=0.3	0.0817***	0.075	0.088	[0.000]
		tau=0.4	0.0864***	0.080	0.093	[0.000]
		tau=0.5	0.0910***	0.085	0.097	[0.000]
		tau=0.6	0.0925***	0.087	0.098	[0.000]
		tau=0.7	0.0938***	0.087	0.101	[0.000]
		tau=0.8	0.0966***	0.089	0.105	[0.000]

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.23: Results of samples grouped by R&D practice

Dependent Variable	Sample	Quantile	Coef.	Conf. lwr	Conf. upr	p-value
R&D expenditure	Total	tau=0.2	-1.31***	-1.360	-1.250	[0.000]
		tau=0.3	-1.30***	-1.340	-1.260	[0.000]
		tau=0.4	-1.29***	-1.320	-1.250	[0.000]
		tau=0.5	-1.29***	-1.330	-1.250	[0.000]
		tau=0.6	-1.30***	-1.340	-1.250	[0.000]
		tau=0.7	-1.27***	-1.320	-1.230	[0.000]
		tau=0.8	-1.27***	-1.320	-1.220	[0.000]
GDPpc_growth	Low expenditure	tau=0.2	-0.199***	-0.217	-0.182	[0.000]
		tau=0.3	-0.197***	-0.213	-0.182	[0.000]
		tau=0.4	-0.194***	-0.206	-0.181	[0.000]
		tau=0.5	-0.190***	-0.204	-0.177	[0.000]
		tau=0.6	-0.192***	-0.205	-0.179	[0.000]
		tau=0.7	-0.191***	-0.202	-0.181	[0.000]
		tau=0.8	-0.196***	-0.216	-0.177	[0.000]
	High expenditure	tau=0.2	0.092***	0.080	0.105	[0.000]
		tau=0.3	0.101***	0.088	0.113	[0.000]
		tau=0.4	0.103***	0.091	0.114	[0.000]
		tau=0.5	0.105***	0.095	0.115	[0.000]
		tau=0.6	0.105***	0.095	0.114	[0.000]
		tau=0.7	0.110***	0.099	0.120	[0.000]
		tau=0.8	0.117***	0.101	0.133	[0.000]

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.24: Results of samples grouped by R&D expenditures

Dependent Variable	Sample	Quantile	Coef.	Conf. lwr	Conf. upr	p-value
R&D practice	Total	tau=0.2	-5.57***	-5.630	-5.510	[0.000]
		tau=0.3	-5.58***	-5.630	-5.530	[0.000]
		tau=0.4	-5.61***	-5.660	-5.560	[0.000]
		tau=0.5	-5.67***	-5.710	-5.630	[0.000]
		tau=0.6	-5.72***	-5.780	-5.670	[0.000]
		tau=0.7	-5.74***	-5.810	-5.680	[0.000]
		tau=0.8	-5.79***	-5.860	-5.730	[0.000]
GDPpc_growth	Low practice	tau=0.2	-0.266***	-0.279	-0.253	[0.000]
		tau=0.3	-0.267***	-0.277	-0.256	[0.000]
		tau=0.4	-0.266***	-0.277	-0.256	[0.000]
		tau=0.5	-0.270***	-0.282	-0.259	[0.000]
		tau=0.6	-0.272***	-0.283	-0.260	[0.000]
		tau=0.7	-0.278***	-0.290	-0.266	[0.000]
		tau=0.8	-0.284***	-0.300	-0.268	[0.000]
	High practice	tau=0.2	0.099***	0.092	0.107	[0.000]
		tau=0.3	0.105***	0.097	0.113	[0.000]
		tau=0.4	0.110***	0.103	0.116	[0.000]
		tau=0.5	0.111***	0.105	0.118	[0.000]
		tau=0.6	0.112***	0.106	0.119	[0.000]
		tau=0.7	0.114***	0.107	0.122	[0.000]
		tau=0.8	0.115***	0.107	0.124	[0.000]

*, **, *** denote p<0.1, <0.05, <0.01.

D.2 Results of Heterogeneous Analyses

Table S.25: Results of samples grouped by geographical or ethno-cultural region

Sample	Quantile	Coef.	Conf. lwr	Conf. upr	p-value
ASEAN	tau=0.2	-0.499***	-0.521	-0.477	[0.000]
	tau=0.3	-0.476***	-0.532	-0.421	[0.000]
	tau=0.4	-0.467***	-0.502	-0.432	[0.000]
	tau=0.5	-0.472***	-0.501	-0.443	[0.000]
	tau=0.6	-0.481***	-0.517	-0.445	[0.000]
	tau=0.7	-0.469***	-0.519	-0.420	[0.000]
	tau=0.8	-0.445***	-0.486	-0.404	[0.000]
West Asia	tau=0.2	-0.787***	-0.821	-0.753	[0.000]
	tau=0.3	-0.768***	-0.808	-0.728	[0.000]
	tau=0.4	-0.768***	-0.797	-0.739	[0.000]
	tau=0.5	-0.774***	-0.800	-0.748	[0.000]
	tau=0.6	-0.768***	-0.792	-0.744	[0.000]
	tau=0.7	-0.763***	-0.788	-0.738	[0.000]
	tau=0.8	-0.747***	-0.787	-0.706	[0.000]
South Asia	tau=0.2	-2.4***	-2.440	-2.360	[0.000]
	tau=0.3	-2.41***	-2.470	-2.350	[0.000]
	tau=0.4	-2.41***	-2.500	-2.320	[0.000]
	tau=0.5	-2.43***	-2.530	-2.330	[0.000]
	tau=0.6	-2.44***	-2.540	-2.330	[0.000]
	tau=0.7	-2.45***	-2.530	-2.380	[0.000]
	tau=0.8	-2.45***	-2.520	-2.380	[0.000]
CIS	tau=0.2	-7.5***	-7.570	-7.440	[0.000]
	tau=0.3	-7.49***	-7.570	-7.420	[0.000]
	tau=0.4	-7.52***	-7.590	-7.450	[0.000]
	tau=0.5	-7.52***	-7.590	-7.460	[0.000]
	tau=0.6	-7.54***	-7.610	-7.480	[0.000]
	tau=0.7	-7.51***	-7.570	-7.450	[0.000]
	tau=0.8	-7.47***	-7.580	-7.370	[0.000]
Europe	tau=0.2	1.67***	1.650	1.700	[0.000]
	tau=0.3	1.68***	1.660	1.700	[0.000]
	tau=0.4	1.67***	1.650	1.690	[0.000]
	tau=0.5	1.67***	1.650	1.690	[0.000]
	tau=0.6	1.67***	1.650	1.690	[0.000]
	tau=0.7	1.67***	1.650	1.700	[0.000]
	tau=0.8	1.69***	1.660	1.720	[0.000]

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.26: Results of samples grouped by “expiration” of patents

Sample	Quantile	Coef.	Conf. lwr	Conf. upr	p-value
	tau=0.2	-0.891***	-0.910	-0.873	[0.000]
	tau=0.3	-0.886***	-0.900	-0.872	[0.000]
	tau=0.4	-0.891***	-0.901	-0.881	[0.000]
Long term	tau=0.5	-0.897***	-0.908	-0.885	[0.000]
	tau=0.6	-0.899***	-0.911	-0.887	[0.000]
	tau=0.7	-0.899***	-0.913	-0.886	[0.000]
	tau=0.8	-0.895***	-0.911	-0.878	[0.000]
	tau=0.2	-0.977***	-0.990	-0.965	[0.000]
	tau=0.3	-0.973***	-0.984	-0.962	[0.000]
	tau=0.4	-0.975***	-0.985	-0.964	[0.000]
15 Years	tau=0.5	-0.975***	-0.986	-0.964	[0.000]
	tau=0.6	-0.974***	-0.985	-0.963	[0.000]
	tau=0.7	-0.974***	-0.985	-0.964	[0.000]
	tau=0.8	-0.971***	-0.983	-0.960	[0.000]
	tau=0.2	-1.33***	-1.350	-1.320	[0.000]
	tau=0.3	-1.33***	-1.340	-1.320	[0.000]
	tau=0.4	-1.33***	-1.340	-1.320	[0.000]
10 Years	tau=0.5	-1.33***	-1.340	-1.320	[0.000]
	tau=0.6	-1.33***	-1.340	-1.320	[0.000]
	tau=0.7	-1.33***	-1.340	-1.320	[0.000]
	tau=0.8	-1.33***	-1.340	-1.320	[0.000]
	tau=0.2	-1.54***	-1.550	-1.520	[0.000]
	tau=0.3	-1.53***	-1.540	-1.520	[0.000]
	tau=0.4	-1.53***	-1.540	-1.520	[0.000]
5 Years	tau=0.5	-1.53***	-1.540	-1.520	[0.000]
	tau=0.6	-1.53***	-1.540	-1.520	[0.000]
	tau=0.7	-1.53***	-1.550	-1.520	[0.000]
	tau=0.8	-1.53***	-1.540	-1.520	[0.000]

*, **, *** denote $p<0.1$, <0.05, <0.01.

Table S.27: Industries in OECD ICIO table.

Industry	ISIC Rev.4	Industry code
Agriculture, hunting, forestry	01, 02	01T02
Fishing and aquaculture	03	03
Mining and quarrying, energy producing products	05, 06	05T06
Mining and quarrying, non-energy producing products	07, 08	07T08
Mining support service activities	09	09
Food products, beverages and tobacco	10, 11, 12	10T12
Textiles, textile products, leather and footwear	13, 14, 15	13T15
Wood and products of wood and cork	16	16
Paper products and printing	17, 18	17T18
Coke and refined petroleum products	19	19
Chemical and chemical products	20	20
Pharmaceuticals, medicinal chemical and botanical products	21	21
Rubber and plastics products	22	22
Other non-metallic mineral products	23	23
Basic metals	24	24
Fabricated metal products	25	25
Computer, electronic and optical equipment	26	26
Electrical equipment	27	27
Machinery and equipment, nec	28	28
Motor vehicles, trailers and semi-trailers	29	29
Other transport equipment	30	30
Manufacturing nec; repair and installation of machinery and equipment	31, 32, 33	31T33
Electricity, gas, steam and air conditioning supply	35	35
Water supply; sewerage, waste management and remediation activities	36, 37, 38, 39	36T39
Construction	41, 42, 43	41T43
Wholesale and retail trade; repair of motor vehicles	45, 46, 47	45T47
Land transport and transport via pipelines	49	49
Water transport	50	50
Air transport	51	51
Warehousing and support activities for transportation	52	52
Postal and courier activities	53	53
Accommodation and food service activities	55, 56	55T56
Publishing, audiovisual and broadcasting activities	58, 59, 60	58T60
Telecommunications	61	61
IT and other information services	62, 63	62T63
Financial and insurance activities	64, 65, 66	64T66
Real estate activities	68	68
Professional, scientific and technical activities	69 to 75	69T75
Administrative and support services	77 to 82	77T82
Public administration and defence; compulsory social security	84	84
Education	85	85
Human health and social work activities	86, 87, 88	86T88
Arts, entertainment and recreation	90, 91, 92, 93	90T93
Other service activities	94,95, 96	94T96
Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use	97, 98	97T98

Appendix E Moderating role of the "Belt and Road" Initiative

E.1 Compared with Global Sample

Table S.28: 97.5% confidence intervals of two samples

Quantile	"Belt and Road" sample				Global sample			
	Coef.	Conf. lwr	Conf. upr	p-value	Coef.	Conf. lwr	Conf. upr	p-value
tau=0.2	-0.894***	-0.908	-0.879	[0.000]	-0.0378***	-0.044	-0.031	[0.000]
tau=0.3	-0.888***	-0.896	-0.880	[0.000]	-0.0347***	-0.041	-0.029	[0.000]
tau=0.4	-0.889***	-0.900	-0.879	[0.000]	-0.0320***	-0.038	-0.026	[0.000]
tau=0.5	-0.890***	-0.901	-0.879	[0.000]	-0.0305***	-0.036	-0.025	[0.000]
tau=0.6	-0.894***	-0.905	-0.883	[0.000]	-0.0316***	-0.037	-0.026	[0.000]
tau=0.7	-0.893***	-0.904	-0.882	[0.000]	-0.0304***	-0.035	-0.025	[0.000]
tau=0.8	-0.895***	-0.905	-0.885	[0.000]	-0.0303***	-0.038	-0.022	[0.000]

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.29: Results of covariates imbalance tests (Dependent variable: GDPpc_growth).

Variable	Overall Mean (St. Dev.)	Subsample		p value (t-test)
		"Belt and Road" Mean (St. Dev.)	Non-"Belt and Road" Mean (St. Dev.)	
(Obs.)	804	402	402	
government	0.758 (0.089)	0.758 (0.069)	0.757 (0.106)	[0.844]
resource	0.030 (0.039)	0.030 (0.040)	0.031 (0.038)	[0.630]
capital	0.238 (0.054)	0.235 (0.046)	0.240 (0.060)	[0.183]
industry	0.265 (0.063)	0.263 (0.048)	0.266 (0.075)	[0.524]
urbanize	0.012 (0.012)	0.013 (0.009)	0.011 (0.015)	[0.068]
EI	0.246 (0.054)	0.242 (0.049)	0.249 (0.058)	[0.052]
open	0.691 (0.450)	0.666 (0.370)	0.717 (0.517)	[0.109]
employment	0.546 (0.085)	0.550 (0.084)	0.542 (0.086)	[0.207]
FDI	0.081 (0.314)	0.086 (0.355)	0.076 (0.266)	[0.646]
reserve	0.109 (0.133)	0.108 (0.148)	0.109 (0.115)	[0.936]
tiva	0.012 (0.011)	0.013 (0.011)	0.012 (0.012)	[0.210]
coop	3.653 (2.507)	3.637 (2.435)	3.670 (2.580)	[0.854]

Table S.30: Results of covariates imbalance tests (Dependent variable: GNIpc_growth).

Variable	Overall Mean (St. Dev.)	Subsample		p value (t-test)
		"Belt and Road" Mean (St. Dev.)	Non-"Belt and Road" Mean (St. Dev.)	
(Obs.)	824	412	412	
government	0.757 (0.091)	0.758 (0.068)	0.755 (0.110)	[0.629]
resource	0.030 (0.039)	0.030 (0.040)	0.031 (0.038)	[0.854]
capital	0.239 (0.056)	0.235 (0.046)	0.243 (0.065)	[0.036]
industry	0.267 (0.067)	0.264 (0.048)	0.271 (0.081)	[0.090]
urbanize	0.012 (0.013)	0.013 (0.009)	0.012 (0.016)	[0.201]
EI	0.245 (0.053)	0.242 (0.049)	0.249 (0.057)	[0.057]
open	0.683 (0.445)	0.664 (0.368)	0.701 (0.510)	[0.232]
employment	0.547 (0.086)	0.550 (0.084)	0.543 (0.088)	[0.248]
FDI	0.081 (0.310)	0.086 (0.351)	0.076 (0.263)	[0.657]
reserve	0.109 (0.132)	0.108 (0.147)	0.111 (0.115)	[0.728]
tiva	0.012 (0.011)	0.012 (0.011)	0.011 (0.012)	[0.138]
coop	3.592 (2.513)	3.611 (2.428)	3.573 (2.598)	[0.832]

Table S.31: Results of covariates imbalance tests (Dependent variable: household consumption).

Variable	Overall Mean (St. Dev.)	Subsample		p value (t-test)
		"Belt and Road" Mean (St. Dev.)	Non-"Belt and Road" Mean (St. Dev.)	
(Obs.)	782	391	391	
government	0.757 (0.091)	0.760 (0.067)	0.754 (0.111)	[0.347]
resource	0.030 (0.037)	0.028 (0.034)	0.032 (0.040)	[0.224]
capital	0.238 (0.054)	0.233 (0.045)	0.243 (0.062)	[0.008]
industry	0.268 (0.070)	0.262 (0.046)	0.274 (0.087)	[0.015]
urbanize	0.012 (0.012)	0.013 (0.008)	0.012 (0.015)	[0.433]
EI	0.246 (0.052)	0.244 (0.047)	0.248 (0.056)	[0.266]
open	0.670 (0.431)	0.653 (0.351)	0.686 (0.498)	[0.280]
employment	0.544 (0.086)	0.545 (0.083)	0.543 (0.088)	[0.722]
FDI	0.083 (0.314)	0.088 (0.354)	0.078 (0.269)	[0.658]
reserve	0.107 (0.130)	0.104 (0.145)	0.109 (0.112)	[0.585]
tiva	0.011 (0.010)	0.011 (0.009)	0.011 (0.011)	[0.671]
coop	3.512 (2.510)	3.469 (2.439)	3.555 (2.582)	[0.631]

Table S.32: Results of covariates imbalance tests (Dependent variable: CO_2 emission based on consumption).

Variable	Overall Mean (St. Dev.)	Subsample "Belt and Road" Mean (St. Dev.)	Subsample Non-"Belt and Road" Mean (St. Dev.)	p value (t-test)
(Obs.)	816	408	408	
government	0.758 (0.092)	0.759 (0.069)	0.756 (0.111)	[0.655]
resource	0.030 (0.037)	0.029 (0.034)	0.032 (0.039)	[0.228]
capital	0.237 (0.054)	0.233 (0.045)	0.241 (0.061)	[0.058]
industry	0.267 (0.068)	0.263 (0.046)	0.271 (0.084)	[0.094]
urbanize	0.013 (0.013)	0.013 (0.009)	0.013 (0.016)	[0.879]
EI	0.246 (0.053)	0.244 (0.048)	0.247 (0.058)	[0.405]
open	0.676 (0.438)	0.652 (0.354)	0.700 (0.508)	[0.120]
employment	0.542 (0.082)	0.547 (0.079)	0.537 (0.085)	[0.078]
FDI	0.085 (0.316)	0.089 (0.356)	0.081 (0.270)	[0.720]
reserve	0.105 (0.127)	0.101 (0.139)	0.109 (0.113)	[0.372]
tiva	0.012 (0.010)	0.012 (0.009)	0.011 (0.012)	[0.839]
coop	3.616 (2.532)	3.518 (2.481)	3.714 (2.582)	[0.270]

Table S.33: Results of covariates imbalance tests (Dependent variable: Comprehensive Wealth per capita).

Variable	Overall Mean (St. Dev.)	Subsample "Belt and Road" Mean (St. Dev.)	Subsample Non-"Belt and Road" Mean (St. Dev.)	p value (t-test)
(Obs.)	652	326	326	
government	0.751 (0.094)	0.755 (0.070)	0.748 (0.113)	[0.368]
resource	0.032 (0.039)	0.029 (0.035)	0.034 (0.042)	[0.085]
capital	0.241 (0.055)	0.237 (0.047)	0.245 (0.063)	[0.044]
industry	0.270 (0.066)	0.266 (0.047)	0.275 (0.081)	[0.085]
urbanize	0.012 (0.012)	0.012 (0.008)	0.012 (0.015)	[0.902]
EI	0.246 (0.051)	0.245 (0.046)	0.247 (0.056)	[0.701]
open	0.694 (0.442)	0.675 (0.345)	0.714 (0.521)	[0.258]
employment	0.543 (0.088)	0.542 (0.086)	0.543 (0.090)	[0.958]
FDI	0.044 (0.078)	0.044 (0.071)	0.044 (0.084)	[0.953]
reserve	0.111 (0.137)	0.105 (0.151)	0.116 (0.122)	[0.340]
tiva	0.012 (0.011)	0.012 (0.010)	0.012 (0.012)	[0.765]
coop	3.853 (2.537)	3.860 (2.439)	3.847 (2.636)	[0.948]

Table S.34: Results of covariates imbalance tests (Dependent variable: Human Capital per capita).

Variable	Overall Mean (St. Dev.)	Subsample: "Belt and Road" Mean (St. Dev.)	Subsample: Non-"Belt and Road" Mean (St. Dev.)	p value (t-test)
(Obs.)	624	312	312	
government	0.754 (0.096)	0.753 (0.073)	0.754 (0.114)	[0.881]
resource	0.031 (0.042)	0.030 (0.042)	0.033 (0.041)	[0.337]
capital	0.240 (0.056)	0.237 (0.048)	0.242 (0.062)	[0.268]
industry	0.267 (0.062)	0.266 (0.047)	0.267 (0.075)	[0.857]
urbanize	0.012 (0.012)	0.012 (0.008)	0.012 (0.015)	[0.977]
EI	0.247 (0.052)	0.245 (0.048)	0.249 (0.056)	[0.364]
open	0.688 (0.445)	0.661 (0.346)	0.714 (0.526)	[0.135]
employment	0.538 (0.087)	0.539 (0.083)	0.536 (0.092)	[0.734]
FDI	0.045 (0.079)	0.042 (0.066)	0.047 (0.090)	[0.482]
reserve	0.112 (0.141)	0.106 (0.154)	0.117 (0.125)	[0.344]
tiva	0.012 (0.011)	0.012 (0.009)	0.012 (0.012)	[0.605]
coop	3.815 (2.543)	3.895 (2.449)	3.734 (2.634)	[0.428]

Table S.35: Results of covariates imbalance tests (Dependent variable: nighttime light).

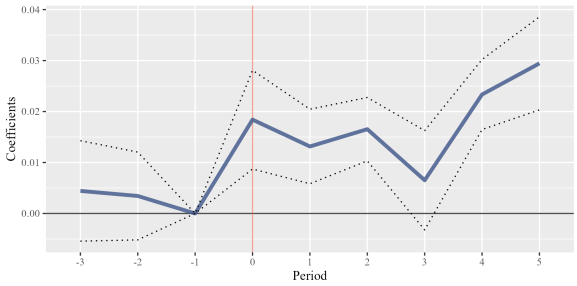
Variable	Overall Mean (St. Dev.)	Subsample: "Belt and Road" Mean (St. Dev.)	Subsample: Non-"Belt and Road" Mean (St. Dev.)	p value (t-test)
(Obs.)	710	355	355	
government	0.764 (0.083)	0.762 (0.062)	0.765 (0.100)	[0.643]
resource	0.029 (0.036)	0.028 (0.036)	0.030 (0.037)	[0.604]
capital	0.238 (0.055)	0.236 (0.045)	0.241 (0.064)	[0.207]
industry	0.269 (0.063)	0.267 (0.046)	0.271 (0.077)	[0.411]
urbanize	0.013 (0.012)	0.013 (0.009)	0.012 (0.014)	[0.535]
EI	0.249 (0.052)	0.248 (0.049)	0.249 (0.056)	[0.797]
open	0.651 (0.382)	0.643 (0.349)	0.658 (0.413)	[0.604]
employment	0.539 (0.085)	0.539 (0.084)	0.538 (0.087)	[0.807]
FDI	0.068 (0.260)	0.071 (0.283)	0.064 (0.237)	[0.755]
reserve	0.094 (0.100)	0.092 (0.114)	0.096 (0.085)	[0.656]
tiva	0.011 (0.010)	0.011 (0.009)	0.011 (0.011)	[0.794]
coop	3.465 (2.485)	3.357 (2.503)	3.572 (2.465)	[0.248]

Table S.36: Results of covariates imbalance tests (Dependent variable: electricity production.

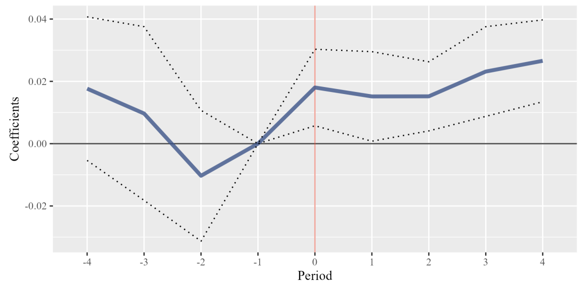
Variable	Overall Mean (St. Dev.)	Subsample "Belt and Road" Mean (St. Dev.)	Subsample Non-"Belt and Road" Mean (St. Dev.)	p value (t-test)
(Obs.)	824	412	412	
government	0.757 (0.091)	0.758 (0.068)	0.755 (0.110)	[0.629]
resource	0.030 (0.039)	0.030 (0.040)	0.031 (0.038)	[0.854]
capital	0.239 (0.056)	0.235 (0.046)	0.243 (0.065)	[0.036]
industry	0.267 (0.067)	0.264 (0.048)	0.271 (0.081)	[0.090]
urbanize	0.012 (0.013)	0.013 (0.009)	0.012 (0.016)	[0.201]
EI	0.245 (0.053)	0.242 (0.049)	0.249 (0.057)	[0.057]
open	0.683 (0.445)	0.664 (0.368)	0.701 (0.510)	[0.232]
employment	0.547 (0.086)	0.550 (0.084)	0.543 (0.088)	[0.248]
FDI	0.081 (0.310)	0.086 (0.351)	0.076 (0.263)	[0.657]
reserve	0.109 (0.132)	0.108 (0.147)	0.111 (0.115)	[0.728]
tiva	0.012 (0.011)	0.012 (0.011)	0.011 (0.012)	[0.138]
coop	3.592 (2.513)	3.611 (2.428)	3.573 (2.598)	[0.832]

Table S.37: Results of covariates imbalance tests (Dependent variable: rail lines).

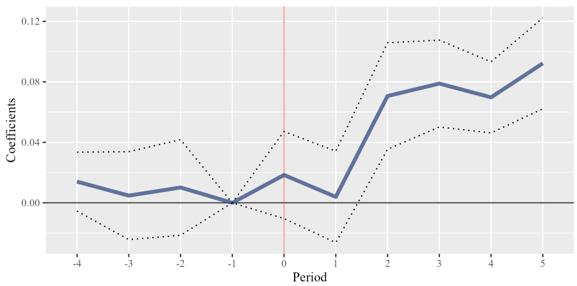
Variable	Overall Mean (St. Dev.)	Subsample "Belt and Road" Mean (St. Dev.)	Subsample Non-"Belt and Road" Mean (St. Dev.)	p value (t-test)
(Obs.)	574	287	287	
government	0.755 (0.083)	0.751 (0.070)	0.759 (0.094)	[0.283]
resource	0.029 (0.041)	0.030 (0.047)	0.028 (0.035)	[0.483]
capital	0.244 (0.058)	0.241 (0.047)	0.247 (0.068)	[0.226]
industry	0.263 (0.061)	0.261 (0.043)	0.265 (0.074)	[0.404]
urbanize	0.011 (0.012)	0.011 (0.009)	0.010 (0.014)	[0.402]
EI	0.257 (0.053)	0.255 (0.047)	0.259 (0.059)	[0.300]
open	0.651 (0.349)	0.649 (0.315)	0.653 (0.382)	[0.907]
employment	0.530 (0.075)	0.533 (0.073)	0.528 (0.078)	[0.348]
FDI	0.044 (0.089)	0.046 (0.086)	0.043 (0.092)	[0.685]
reserve	0.099 (0.118)	0.101 (0.147)	0.096 (0.080)	[0.620]
tiva	0.011 (0.010)	0.011 (0.010)	0.010 (0.010)	[0.335]
coop	3.439 (2.494)	3.363 (2.473)	3.514 (2.517)	[0.466]



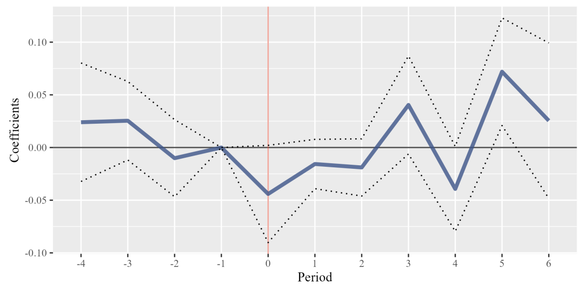
(a)



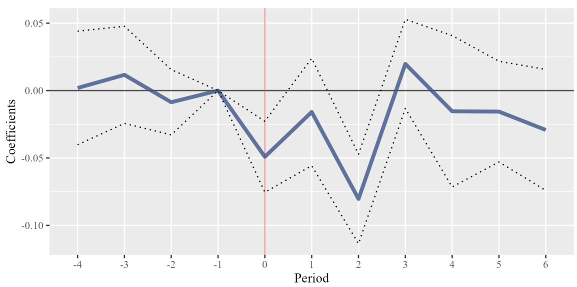
(b)



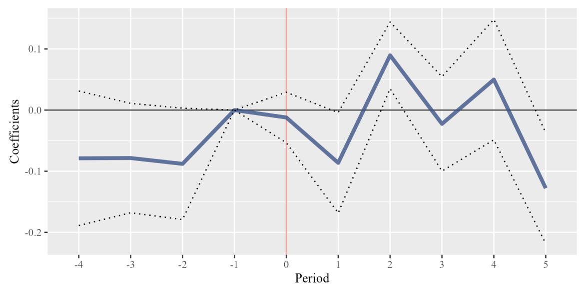
(c)



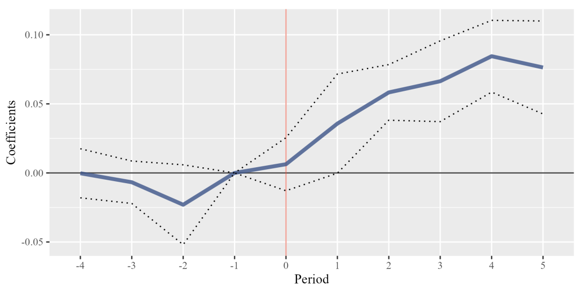
(d)



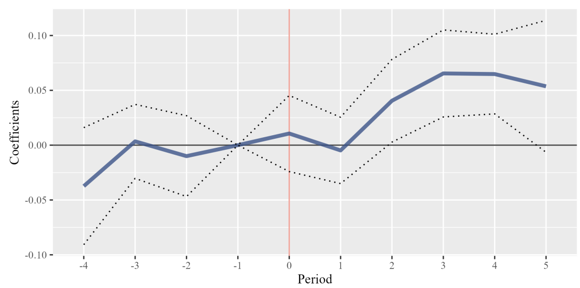
(e)



(f)



(g)



(h)

Figure S.2: Results of Parallel tests.

Note: NOTE: Subfigure (a) shows parallel tests results of PSM sample with GNIpc_growth as the dependent variable; Subfigure (b) shows the results of household consumption; Subfigure (c) shows the results of CO_2 emission based on consumption; Subfigure (d) shows the results of Comprehensive Wealth per capita; Subfigure (e) shows the results of Human Capital per capita; Subfigure (f) shows the results of nighttime light; Subfigure (g) shows the results of electricity production; Subfigure (h) shows the results of rail lines.

Table S.38: Moderating effect results of Belt and Road with different aspects of economic development.

Variable	Economic Growth						
	(1) tau=0.2	(2) tau=0.3	(3) tau=0.4	(4) tau=0.5	(5) tau=0.6	(6) tau=0.7	(7) tau=0.8
Panel A: GNIpc_growth ∼							
GPYC_rank	−0.022***	−0.024***	−0.022***	−0.017***	−0.016***	−0.015***	−0.013***
	(0.006)	(0.005)	(0.006)	(0.005)	(0.004)	(0.005)	(0.005)
BRI	−0.008**	−0.002	−0.001	0.005	0.003	0.004	0.002
	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)	(0.003)	(0.004)
GPYC_rank×BRI	−0.019***	−0.022***	−0.022**	−0.031***	−0.026***	−0.026***	−0.017**
	(0.007)	(0.008)	(0.009)	(0.007)	(0.007)	(0.008)	(0.008)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	662	662	662	662	662	662	662
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.494	0.512	0.516	0.517	0.520	0.518	0.510
Panel B: household consumption ∼							
GPYC_rank	0.174***	0.175***	0.179***	0.179***	0.176***	0.178***	0.163***
	(0.009)	(0.009)	(0.008)	(0.007)	(0.006)	(0.006)	(0.010)
BRI	−0.006	−0.003	0.008	0.008	0.007	0.010*	0.011*
	(0.006)	(0.007)	(0.007)	(0.006)	(0.005)	(0.006)	(0.005)
GPYC_rank×BRI	−0.380***	−0.381***	−0.395***	−0.387***	−0.383***	−0.383***	−0.378***
	(0.011)	(0.013)	(0.014)	(0.013)	(0.010)	(0.011)	(0.013)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	613	613	613	613	613	613	613
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.708	0.716	0.718	0.718	0.712	0.703	0.690
Panel C: CO_2 emission based on consumption ∼							
GPYC_rank	−0.582***	−0.588***	−0.572***	−0.580***	−0.566***	−0.570***	−0.566***
	(0.028)	(0.034)	(0.026)	(0.022)	(0.024)	(0.028)	(0.027)
BRI	−0.014	−0.031	−0.002	0.002	0.002	0.003	0.005
	(0.022)	(0.020)	(0.021)	(0.016)	(0.014)	(0.015)	(0.017)
GPYC_rank×BRI	−0.182***	−0.152***	−0.208***	−0.193***	−0.195***	−0.202***	−0.211***
	(0.043)	(0.042)	(0.041)	(0.035)	(0.031)	(0.029)	(0.033)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	656	656	656	656	656	656	656
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.701	0.701	0.705	0.712	0.718	0.719	0.717
Panel D: Comprehensive Wealth per capita ∼							
GPYC_rank	−0.444***	−0.444***	−0.449***	−0.452***	−0.467***	−0.442***	−0.441***
	(0.026)	(0.022)	(0.018)	(0.017)	(0.025)	(0.024)	(0.026)
BRI	0.000	0.011	0.002	−0.004	−0.010	−0.011	0.000
	(0.016)	(0.014)	(0.014)	(0.012)	(0.012)	(0.013)	(0.017)
GPYC_rank×BRI	−1.042***	−1.049***	−1.007***	−0.982***	−0.963***	−0.961***	−0.982***
	(0.047)	(0.038)	(0.036)	(0.028)	(0.033)	(0.035)	(0.038)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	546	546	546	546	546	546	546
Year FE	✓	✓	✓	✓	✓	✓	✓

	(1) tau=0.2	(2) tau=0.3	(3) tau=0.4	(4) tau=0.5	(5) tau=0.6	(6) tau=0.7	(7) tau=0.8
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.813	0.797	0.778	0.757	0.735	0.710	0.681
Panel E: Human Capital per capita $\sim$							
GPYC_rank	−0.123***	−0.121***	−0.116***	−0.102***	−0.114***	−0.108***	−0.105***
	(0.026)	(0.024)	(0.026)	(0.022)	(0.016)	(0.020)	(0.030)
BRI	0.957***	0.956***	0.961***	0.963***	0.961***	0.948***	0.939***
	(0.020)	(0.013)	(0.014)	(0.013)	(0.012)	(0.013)	(0.019)
GPYC_rank×BRI	−0.492***	−0.493***	−0.494***	−0.498***	−0.496***	−0.448***	−0.407***
	(0.045)	(0.032)	(0.033)	(0.029)	(0.033)	(0.043)	(0.039)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	519	519	519	519	519	519	519
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.772	0.803	0.826	0.843	0.841	0.830	0.816
Panel F: nighttime light $\sim$							
GPYC_rank	−2.260***	−2.246***	−2.223***	−2.210***	−2.191***	−2.155***	−2.110***
	(0.054)	(0.031)	(0.023)	(0.027)	(0.036)	(0.039)	(0.051)
BRI	−3.565***	−3.575***	−3.551***	−3.568***	−3.566***	−3.557***	−3.553***
	(0.038)	(0.026)	(0.023)	(0.023)	(0.019)	(0.020)	(0.021)
GPYC_rank×BRI	−0.611***	−0.531***	−0.564***	−0.527***	−0.518***	−0.539***	−0.527***
	(0.082)	(0.056)	(0.043)	(0.047)	(0.041)	(0.041)	(0.042)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	527	527	527	527	527	527	527
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.918	0.931	0.942	0.951	0.945	0.935	0.920
Panel G: electricity production $\sim$							
GPYC_rank	0.387***	0.380***	0.370***	0.357***	0.323***	0.334***	0.345***
	(0.021)	(0.018)	(0.018)	(0.018)	(0.017)	(0.020)	(0.026)
BRI	0.000	0.007	0.007	0.001	−0.002	0.012	0.024*
	(0.016)	(0.012)	(0.012)	(0.012)	(0.013)	(0.015)	(0.013)
GPYC_rank×BRI	−0.406***	−0.427***	−0.417***	−0.408***	−0.396***	−0.411***	−0.424***
	(0.026)	(0.027)	(0.024)	(0.023)	(0.025)	(0.033)	(0.032)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	592	592	592	592	592	592	592
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.741	0.746	0.751	0.747	0.744	0.739	0.737
Panel H: rail lines $\sim$							
GPYC_rank	−2.260***	−2.246***	−2.223***	−2.210***	−2.191***	−2.155***	−2.110***
	(0.054)	(0.031)	(0.023)	(0.027)	(0.036)	(0.039)	(0.051)
BRI	1.905***	1.900***	1.913***	1.917***	1.927***	1.939***	1.946***
	(0.020)	(0.014)	(0.012)	(0.013)	(0.015)	(0.014)	(0.017)
GPYC_rank×BRI	−2.378***	−2.392***	−2.418***	−2.424***	−2.437***	−2.480***	−2.515***
	(0.055)	(0.036)	(0.028)	(0.031)	(0.040)	(0.041)	(0.053)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	406	406	406	406	406	406	406
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.921	0.923	0.926	0.929	0.932	0.935	0.939

	(1) tau=0.2	(2) tau=0.3	(3) tau=0.4	(4) tau=0.5	(5) tau=0.6	(6) tau=0.7	(7) tau=0.8

*, **, *** denote p<0.1, <0.05, <0.01.

E.2 Interaction with External Factors

Table S.39: Results of moderating effect interacting with Ideal Point.

Variable	GDPpc_growth						
	(1) tau=0.2	(2) tau=0.3	(3) tau=0.4	(4) tau=0.5	(5) tau=0.6	(6) tau=0.7	(7) tau=0.8
Panel A: ∼ Ideal Point							
GPYC_rank	-0.107***	-0.106***	-0.103***	-0.105***	-0.105***	-0.107***	-0.104***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
ideal	0.005***	0.003***	0.003***	0.001	-0.001	-0.002**	-0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
GPYC_rank×ideal	-0.014***	-0.011***	-0.007***	-0.002	0.000	0.001	0.006**
	(0.003)	(0.003)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	1506	1506	1506	1506	1506	1506	1506
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.543	0.554	0.559	0.562	0.563	0.557	0.546
Panel B: ∼ Ideal Point & Belt and Road							
GPYC_rank	-0.128***	-0.122***	-0.118***	-0.117***	-0.117***	-0.115***	-0.115***
	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)
ideal	-0.003*	-0.003**	-0.004**	-0.005***	-0.006***	-0.006***	-0.008***
	(0.002)	(0.001)	(0.002)	(0.001)	(0.002)	(0.001)	(0.001)
GPYC_rank×ideal	0.003	0.002	0.007*	0.007**	0.008**	0.007**	0.014***
	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)	(0.003)	(0.003)
GPYC_rank×ideal×BRI	-0.061***	-0.049***	-0.053***	-0.044***	-0.043***	-0.037***	-0.043***
	(0.006)	(0.007)	(0.007)	(0.006)	(0.007)	(0.007)	(0.007)
BRI	-0.008**	-0.003	0.000	0.004	0.004	0.007**	0.007*
	(0.004)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
GPYC_rank×BRI	-0.016**	-0.018***	-0.024***	-0.029***	-0.028***	-0.030***	-0.033***
	(0.007)	(0.007)	(0.006)	(0.006)	(0.006)	(0.006)	(0.007)
ideal×BRI	0.025***	0.021***	0.022***	0.018***	0.017***	0.016***	0.019***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	1506	1506	1506	1506	1506	1506	1506
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.564	0.570	0.573	0.576	0.578	0.576	0.566

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.40: Results of moderating effect interacting with geopolitical risk.

Variable	GDPpc_growth						
	(1) tau=0.2	(2) tau=0.3	(3) tau=0.4	(4) tau=0.5	(5) tau=0.6	(6) tau=0.7	(7) tau=0.8
Panel A: ∼ Geopolitical Risk							
GPYC_rank	-0.039***	-0.035***	-0.042***	-0.046***	-0.047***	-0.056***	-0.057***
	(0.010)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.008)
geopolit	0.058***	0.058***	0.049***	0.043***	0.042***	0.035***	0.029***
	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)	(0.005)
GPYC_rank×geopolit	-0.122***	-0.123***	-0.111***	-0.103***	-0.103***	-0.089***	-0.085***
	(0.012)	(0.009)	(0.008)	(0.009)	(0.008)	(0.008)	(0.010)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	1770	1770	1770	1770	1770	1770	1770
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.566	0.581	0.592	0.599	0.601	0.595	0.580
Panel B: ∼ Geopolitical Risk & Belt and Road							
GPYC_rank	-0.109***	-0.108***	-0.102***	-0.097***	-0.094***	-0.093***	-0.088***
	(0.016)	(0.010)	(0.009)	(0.011)	(0.012)	(0.013)	(0.015)
geopolit	0.011	0.008	0.010	0.011	0.011	0.007	0.008
	(0.011)	(0.007)	(0.007)	(0.008)	(0.009)	(0.010)	(0.011)
GPYC_rank×geopolit	-0.058***	-0.057***	-0.059***	-0.063***	-0.065***	-0.064***	-0.067***
	(0.017)	(0.012)	(0.012)	(0.013)	(0.013)	(0.014)	(0.016)
GPYC_rank×geopolit×BRI	-0.117***	-0.110***	-0.099***	-0.089***	-0.087***	-0.075***	-0.086***
	(0.029)	(0.016)	(0.021)	(0.019)	(0.020)	(0.021)	(0.024)
BRI	-0.010	-0.012	-0.004	0.003	0.004	0.006	0.003
	(0.014)	(0.007)	(0.009)	(0.009)	(0.011)	(0.012)	(0.012)
GPYC_rank×BRI	0.126***	0.128***	0.118***	0.106***	0.102***	0.089***	0.095***
	(0.022)	(0.012)	(0.016)	(0.014)	(0.016)	(0.016)	(0.018)
geopolit×BRI	0.075***	0.077***	0.068***	0.064***	0.066***	0.068***	0.075***
	(0.017)	(0.010)	(0.012)	(0.011)	(0.013)	(0.014)	(0.015)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	1770	1770	1770	1770	1770	1770	1770
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.676	0.693	0.700	0.696	0.688	0.676	0.657

*, **, *** denote p<0.1, <0.05, <0.01.

Table S.41: Results of moderating effect interacting with Weighted Average Tariff.

Variable	GDPpc_growth						
	(1) tau=0.2	(2) tau=0.3	(3) tau=0.4	(4) tau=0.5	(5) tau=0.6	(6) tau=0.7	(7) tau=0.8
Panel A: ∼ Tariff							
GPYC_rank	-0.177***	-0.179***	-0.175***	-0.177***	-0.174***	-0.176***	-0.169***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
wtar	-0.583***	-0.492***	-0.543***	-0.477***	-0.550***	-0.533***	-0.810***
	(0.055)	(0.075)	(0.093)	(0.083)	(0.101)	(0.145)	(0.132)
GPYC_rank×wtar	1.366***	1.125***	1.175***	1.025***	1.119***	1.042***	1.309***
	(0.163)	(0.164)	(0.157)	(0.149)	(0.157)	(0.183)	(0.207)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	1738	1738	1738	1738	1738	1738	1738
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.665	0.676	0.682	0.687	0.688	0.681	0.666
Panel B: ∼ Tariff & Belt and Road							
GPYC_rank	-0.182***	-0.182***	-0.175***	-0.173***	-0.171***	-0.165***	-0.160***
	(0.004)	(0.004)	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)
wtar	-1.040***	-1.102***	-1.131***	-1.189***	-1.363***	-1.543***	-1.746***
	(0.203)	(0.140)	(0.106)	(0.145)	(0.175)	(0.194)	(0.248)
GPYC_rank×wtar	2.368***	2.276***	2.353***	2.400***	2.627***	2.818***	2.963***
	(0.312)	(0.237)	(0.173)	(0.193)	(0.212)	(0.260)	(0.355)
GPYC_rank×wtar×BRI	-1.537***	-1.925***	-2.238***	-2.155***	-2.247***	-2.595***	-2.429***
	(0.472)	(0.408)	(0.322)	(0.266)	(0.346)	(0.470)	(0.431)
BRI	-0.016***	-0.008**	-0.002	0.001	0.005	0.010**	0.012***
	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)
GPYC_rank×BRI	0.020**	0.010	-0.002	-0.005	-0.008	-0.014	-0.012
	(0.008)	(0.008)	(0.007)	(0.006)	(0.007)	(0.009)	(0.008)
wtar×BRI	0.502**	0.732***	0.866***	0.853***	0.966***	1.220***	1.225***
	(0.229)	(0.161)	(0.193)	(0.199)	(0.237)	(0.307)	(0.306)
Control Variable	✓	✓	✓	✓	✓	✓	✓
Observation	1738	1738	1738	1738	1738	1738	1738
Year FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓
R_1	0.688	0.695	0.699	0.701	0.702	0.695	0.685

*, **, *** denote p<0.1, <0.05, <0.01.

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